Online Resource 1: Supporting Tables

This PDF, `QSA_SQJ_Online_Resource_1.pdf`, accompanies *Diagnosing Undetected Quantum Mutants: Equivalence, Input Contracts and Oracle Statistics.*

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Online Resource 2 is `QSA_SQJ_Online_Resource_2.zip`; all archive paths below are relative to its `package/` root. The small cited JSON/CSV result files are shipped at their stated paths; any generated-only path is explicitly marked as regenerated by reproduce.sh at that path. The tables retain fixed-mutant-set counts and historical outcomes. All later census, test-change, contract and sensitivity analyses are post-hoc and exploratory; the added statistic families were fixed before their own calculations, after exposure to earlier results.

Terminology used in this resource.

A subject is a provided mutant source file; a release control is the corresponding supplied unmutated implementation. A slot fixes circuit arguments or state preparation; the fixed wrapper adds this preparation and differential tests to an archived program, producing a whole circuit, the build. The registered input is the all-zero state before that build; the application input is the state after slot preparation. A measurement basis is an output axis, Z, X or Y; the test catalogue crosses 4 slots with those axes. A test pairs a slot with a measurement basis; the test catalogue contains all tests, and a suite selects a subset. A silent subject is undetected by that entire test catalogue. A mutant set contains the subjects checked together. An input contract specifies allowed inputs and initial register states. The executor is the circuit runner; its executor contract admits all preparable basis strings. The program-semantic contract adds documented source/property restrictions, with the executor contract where none are documented. Workspace registers hold intermediate data; a workspace contract assumes their fixed initialization. The census classifies every subject by cause relative to its input contract. An equivalent build (E) has full-unitary agreement up to global phase and tolerance. An input-limited subject (I) agrees on registered states but differs elsewhere; it aggregates A, C and P. An admissible-input witness (A) has a state difference on an admissible basis input. A contract-limited subject (C) differs only outside the input contract; a phase-only subject (P) agrees on each admissible basis output up to its own phase, while phases differ between inputs. The existing statistics are one-qubit marginals and whole-register parity. A summary-limited subject (O) changes a registered state whose full histogram differs although existing statistics agree. A threshold-limited subject (T) has a resolved positive gap no larger than the threshold; D denotes detected. A numerical state witness

has infidelity above the state tolerance ϵ_s, establishing a state difference, not necessarily a detection by existing statistics (one-qubit marginals and whole-register parity). Measurement-basis-limited (M) means different output states with identical full histograms in the registered measurement bases; that class is 0 at registered inputs here. A *behaviour group* contains subjects from the same program whose unitaries agree up to global phase in every slot within the unitary tolerance ϵ_u; groups do not establish independent failure mechanisms. QuCheck has 77 such groups and Muskit 608. Exact calculations use double-precision state vectors or unitaries and stated tolerances; they are numerical checks rather than symbolic proofs.

Notation and comparison definitions.

Let m index a subject, n its qubit count, ℓ its slot, $b = 0, 1, 2$ its Z/X/Y basis, and j an oracle statistic; test $t = 3\ell + b$. For exact subject/control probabilities p_j, q_j, an exact gap is $|p_j - q_j|$; hats denote shot estimates, whose absolute difference is a saved-shot gap when computed from stored counts. Write $d_{\ell b} = \max_j |p_j - q_j|$ and $d = \max_{\ell,b} d_{\ell b}$; hats denote sampled gaps and every table uses d, not a detection count. For subject/control unitaries $U_{m,\ell}, U_{c,\ell}$ and registered outputs $\psi_{m,\ell}, \psi_{c,\ell}$, define

$$u_\ell = |1 - |\operatorname{tr}(U_{m,\ell}^\dagger U_{c,\ell})|/2^n|, \qquad f_\ell = 1 - |\langle\psi_{m,\ell}|\psi_{c,\ell}\rangle|^2.$$

Here tr is the trace and † conjugate transpose. For unitary overlap deviation u_ℓ and registered-state infidelity f_ℓ, $u = \max_\ell u_\ell$ and $f = \max_\ell f_\ell$; $\epsilon_u, \epsilon_s, \epsilon_o$ are their unitary, state and gap tolerances (Table S3). With $N = 8{,}192$ shots, comparison allowance K and per-subject error allowance $\alpha = 0.001$, the threshold is $\tau(K) = \sqrt{2\ln(4K/\alpha)/N}$; $\tau = \tau_0$ uses $K = 120$. The pairwise parity observable is $P(q_r \oplus q_s = 1)$ for qubits $r < s$; $\oplus$ denotes XOR. Its comparison count is $L = 12[n + 1 + \binom{n}{2}]$, with $\tau_{\text{pair}} = \tau(\max(120, L))$; τ_{TV} denotes the separately adjusted total-variation threshold. Total variation (TV) is $\frac{1}{2}\sum_z |p(z) - q(z)|$ over output bins z. A cell-minimum margin minimizes detected cell gaps then subtracts the stated threshold; a subject-maximum margin first maximizes within each subject, then minimizes across subjects and subtracts that threshold. A lineage groups programs sharing source code; an algorithm family groups implementations of the same algorithm. The historical development set supplies 60 Fourier-transform/phase-estimation subjects for selection; the evaluation set supplies 120 subjects from the other 4 QuCheck algorithms for reporting. Greedy selection starts at test 0, maximizes newly detected development subjects, breaks ties by smallest test ID and continues through zero-gain additions until 4 tests. The inherited suite $T_M = [0, 11, 3, 1]$ was selected on the development set; its baseline $T_B = \{0\}$ is retained by each equal-budget comparator. Budget k counts target test invocations at fixed shots, excluding controls; an invocation runs one target test. Historical formulas use c_{base}, c_{inh} and c_{all} for baseline, inherited-suite and test-catalogue detection counts, n_{sub} for the subject denominator and $\mathcal{S}$ for a suite. The probability gap remains d, the slot index ℓ and the comparison allowance K; $\mathcal{X}$ indexes historical contrasts and v indexes programs. Roman S/B always name fresh-shot arms; M/T name census classes and J counts elementary tests. Here $\mathcal{T}$ is the test catalogue, and $c(\mathcal{S})$ counts subjects detected by at least one test in suite $\mathcal{S}$. The uniform reference

distribution is the detection-count distribution induced by equal weights on comparator suites; the uniform mean averages their detection counts, raw gain subtracts the baseline count, and excess subtracts the uniform mean from the inherited count. The inclusive tail fraction is the proportion of comparator counts at least the inherited count, including ties; it is descriptive, not a population p-value. The test-catalogue union is the union of all test detection sets; the detection matrix contains subject–test decisions. A replay repeats the same inputs, seeds and node; worker elapsed time sums measured worker execution regions, excluding end-to-end costs.

Historical analyses and RQ scope.

Suite selection, equal-budget contrasts, margin comparisons, C1–C4 holdout checks and source-tool outcome cross-tabs describe earlier analyses; none supplies evidence for RQ1 or RQ2. C1–C4 retain their original predicates, implementation limits and verdict codes; "incomplete" means excluded unbuildable subjects lack the planned-denominator analysis.

Resource map.

Census, numerical criteria and budget sweep: Tables S1, S3, S2. Source comparisons and edit inventory: Tables S4, S5, S20. Execution and hashes: Tables S6, S25. Agreement and coverage: Tables S7, S8. Timing and comparator families: Tables S9, S28, S10, S11, S12. Muskit inputs, exclusions, magnitudes and fixed predicates: Tables S13, S14, S15, S16, S17. Admissibility, behaviour groups, boundaries, test changes, validation and witnesses: Tables S18, S19, S21, S22, S23, S24. Exposure chronology and the worked suite example: Tables S29 and S32. Margin nesting and historical reconciliation: Tables S34 and S33. All workspace masks, original cohorts and mixed/higher-weight evidence: Tables S35–S42. Registered Pearson, independent-draw calibration and native attempts: Tables S43–S45. Slot-consistency audit, file/group census and exact-law searches: Tables S46–S50. Fresh-shot search, specificity, source-tool linkage and binary dependencies: Tables S51–S56. Duplicate signatures and 1-test rows: Tables S26, S27.

Original Muskit holdout checks

Muskit was chosen as an available licensed mutant set with release controls, taking eligible static-register programs outside QuCheck's families. Its source rule, slot preparations, inherited suite and descriptive checks were fixed and hashed before Muskit outcomes, after QuCheck exposure. Here the historical *holdout* means that Muskit's observations had not yet been inspected when those choices were fixed; it is not fresh confirmation for the later census or repair analyses. The buildable-subject analysis is complete; the original planned-denominator analysis is incomplete. The test catalogue detects 621/1,136 subjects, and the all-Z suite detects 582. Detection breadth, the number of detecting tests per detected subject, has mean 6.44 and minimum 1. The 2 single-test CE detections are kept in Table S27.

Table S17 states each historical predicate, its direction and its unchanged outcome. C1 asks: do decisions from stored samples agree with exact probabilities, and do

Table S1 Exact census and agreement base. E: equivalent builds at all slots; I: input-limited; O: summary-limited. Silent E/I/O/T columns partition undetected subjects; T is threshold-limited.

Mutant set / program	Files	Detected	Silent E	Silent I	Silent O	Silent T
QuCheck	120	70	41	9	0	0
Muskit pooled	1,136	621	154	311	50	0
CE	446	233	63	100	50	0
LittleBV	333	207	44	82	0	0
QRAM	357	181	47	129	0	0

Implementation agreement / state sensitivity	QuCheck	Muskit
Agreeing mutant–test cells / all cells	1,440/1,440	13,632/13,632
Minimum d above threshold	0.146447	0.066987
Detected margin	0.089933	0.010474
Undetected margin	0.056513	0.006563
Subjects with a registered-state difference	70/120	671/1,136
Registered subject–input cells with a state difference	264/480	2,531/4,544
Summary-limited subject–input cells	0	183
Positive subthreshold subject–test cells (QRAM)	—	20
Subjects contributing those cells (QRAM)	—	5

Detected gap minima here range over detected individual subject–test–statistic cells; margins subtract τ. In Table S21, the D-row gap instead first maximizes over tests and statistics within each subject, then minimizes over subjects. QuCheck cell/subject-max minima are 0.146447/0.250000; Muskit minima are 0.066987/0.103553. Undetected margin: τ minus the maximum undetected gap. Summary-limited subject–input cells count a subject and slot once, across all measurement bases: QuCheck has 0 of 480, and Muskit 183 of 4,544. Threshold: 0.0565. Circuits/controls are shared. Census zero tolerance: 10^{-12}; frozen Muskit agreement tolerance: 10^{-10}.

Positive subthreshold counts project `semantic_summary.json`, selector `catalogues.Muskit.invisible.state_differs_positive_subthreshold_basis` (`cells`, `subjects`); all are QRAM subjects detected in another basis. Cell records: Online Resource 2, `r3/analysis/results/semantic_cells.csv`.

subjects labelled equivalent remain undetected? C2 asks: do tests detect some but not all subjects, do several tests detect each detected subject, and does Z detect most of the detected union? C3 asks: does the mean over equal-budget suites explain at least half the inherited suite's gain, does another suite tie or beat it, and does its rank change with the comparison set? C4 asks: does a positive gain disappear against the uniform mean, or does the sign of that difference change in the specified contrasts? Its historical equivalent-label clause is retained below. The label checks are unavailable because Muskit supplies no equivalent-subject labels. The recorded C2 conjunction is partial. C4 did not replicate: the evaluated pooled and omission comparisons did not give the required nonpositive excess or sign change. The ordinal-transfer outcome is limited to particular lineages, and pooled excess does not extend to every program. These outcomes concern historical descriptive checks and supply no evidence for RQ1 or RQ2. The inclusive fractions 81/165 and 229/495 use baseline-preserving and unconstrained suites respectively, so their denominators and reference sets differ.

Table S2 Budget sweep. Prefix detections use the first k tests of $[0, 11, 3, 1]$. The retained-baseline family retains test 0; the all-suites family is unconstrained. Here k counts tests. Each panel reports its uniform mean, excess over that mean, and inclusive count / all suites.

Scope	k	Prefix detections	Retained mean	Retained excess	Suites ≥ prefix / retained	All mean	All excess	Suites ≥ prefix / all
QuCheck	1	50	50	0	1/1	41.75	8.25	4/12
	2	63	59.36	3.64	4/11	56.97	6.03	19/66
	3	64	64.18	-0.18	37/55	63.52	0.48	131/220
	4	68	66.73	1.27	57/165	66.54	1.46	255/495
Muskit	1	529	529	0	1/1	333.25	195.75	4/12
	2	565	562.82	2.18	3/11	464.09	100.91	30/66
	3	590	580.62	9.38	18/55	529.46	60.54	94/220
	4	596	591.71	4.29	81/165	565.24	30.76	229/495
CE	1	191	191.00	0.00	1/1	118.25	72.75	3/12
	2	213	210.82	2.18	8/11	166.73	46.27	25/66
	3	222	219.51	2.49	20/55	193.46	28.54	67/220
	4	228	223.86	4.14	44/165	209.05	18.95	86/495
LittleBV	1	186	186.00	0.00	1/1	112.50	73.50	4/12
	2	186	189.82	-3.82	11/11	159.32	26.68	38/66
	3	195	193.25	1.75	27/55	180.22	14.78	136/220
	4	195	196.31	-1.31	109/165	190.62	4.38	369/495
QRAM	1	152	152.00	0.00	1/1	102.50	49.50	4/12
	2	166	162.18	3.82	3/11	138.05	27.95	30/66
	3	173	167.85	5.15	12/55	155.78	17.22	77/220
	4	173	171.54	1.46	78/165	165.57	7.43	262/495

Denominators: QuCheck 120 evaluation mutants (60 development mutants); Muskit 1,136, CE 446, LittleBV 333, QRAM 357. Ties and the prefix itself are included. On the separate 60 development subjects, cumulative detections by the selected 1-, 2-, 3- and 4-test prefixes are 18, 38, 39, 39; these are development detections, not evaluation gains. Muskit has no development split. Test index $t = 3\ell + b$ uses Z/X/Y for $b = 0, 1, 2$. Complete distributions: Online Resource 2, `r3/analysis/results/budget_sweep.json`.

Reading the archived verdicts.

The verdict REPLICATION means the stated check holds; NON_REPLICATION means it fails, LINEAGE_LIMITED holds only in named lineages, PARTIAL lacks part of a conjunction, and UNAVAILABLE lacks the required labels. Read C1–C4 in order in the following table: first the predicate, then its evaluated scope, then its recorded verdict. The comparison set is the baseline-preserving suite family unless the row explicitly names the unconstrained contrast; their pooled ranks use different denominators. For C4, distinguish the within-scope sign condition from the separate lineage-dependent ordinal-transfer check; neither supplies a verdict for an unlisted scope. The notation preceding each predicate defines its comparison family and counts, and the final table retains the original directions and outcomes.

Table S3 Numerical criteria. The frozen agreement band does not round or change the strict decision $d > \tau$.

Criterion	Tested quantity	Predicate
Unitary equivalence	Phase-invariant overlap deviation u_ℓ	$u_\ell \leq \epsilon_u = 10^{-9}$
State difference	Infidelity f_ℓ	$f_\ell > \epsilon_s = 10^{-10}$
Observable zero	Probability gap $d_{\ell b}$	$d_{\ell b} \leq \epsilon_o = 10^{-12}$
Frozen Muskit agreement	Gap d and threshold distance	$d \leq 10^{-10}$ (zero);
		$\lvert d - \tau \rvert \leq 10^{-10}$ (borderline)

A borderline cell retains its strict Boolean decision but makes agreement numerically weak under the frozen check. The state criterion and agreement band happen to use the same value; they test different quantities.

Table S4 Closest approaches, their existing mechanisms and evaluations, and the evidence added by this finite-test-catalogue audit.

Approach	Existing mechanism and evaluation	Difference studied here
QuCheck	Property generators; multi-basis circuit assertions; statistical correction	Complete fixed detection matrix, test-catalogue union and exhaustive suite distribution
Quito	Value-based coverage; specified-probability checks; detection and execution cost	Measurement basis as a test-catalogue dimension; executed release controls
MutTG	Joint suite-size and detection search; matched search evaluations	All equally sized alternatives under a baseline-preservation constraint
Fortunato et al. (2022)	Qiskit mutation case study; equivalent mutants and oracle choice	Input-contract and statistic limits separated for supplied builds

Source inventory and inspected records: Online Resource 2, `r3/literature/RELATED_WORK.md`; bibliography: Online Resource 2, `r3/paper/refs.bib`.

Table S5 Source-derived mutant composition. Each row totals 30 files. Preserving is an upstream declaration; diff categories are mutually exclusive after that precedence.

Family	Set	Preserving	Delete	Insert	Replace	Mixed	Total
Deutsch–Jozsa	Evaluation	10	2	8	10	0	30
Grover	Evaluation	10	1	9	10	0	30
Fourier transform	Development	10	2	13	5	0	30
Phase estimation	Development	10	1	6	12	1	30
Teleportation	Evaluation	10	1	9	10	0	30
Superdense coding	Evaluation	10	3	7	10	0	30

Table S6 Execution inventory. QuCheck records are simulator invocations, including repeated controls; target counts differ from unique mutant–test cells. Holdout jobs include controls and retain build failures.

QuCheck stage	Target calls	Control calls	Records
Development	720	72	792
Original evaluation, including baseline repeat	600	60	660
Added candidates	960	96	1,056
Quality-control replay	480	48	528
Sources			186
Mutants			180
Controls			6
Frozen builds			744
Unique mutant–test cells			2,160
Total records			3,036
Shots per invocation			8,192
Muskit holdout quantity			**Count or time**
Planned sources (mutants + controls)			1,175 (1,172 + 3)
Sources excluded by build errors			36
Failed jobs			432
Planned and attempted jobs			14,100
Valid jobs			13,668
Valid mutant jobs			13,632
Valid control jobs			36
Shots per valid job			8,192
Sampled shots			111,968,256
Planned shots			115,507,200
Archived CPU field			221.05

QuCheck's 2,040 evaluation target records sum original, added and replay calls; development is separate. Its 660 before/after comparisons include controls and repeated prior baselines. The holdout CPU value is an archived analyzer accounting field; its timer region and process scope cannot be established here, so it is excluded from timing comparisons; enumeration adds no sampled jobs.

Table S7 Numerical oracle agreement on $120 \times 12 = 1{,}440$ mutant–test cells.

Separate observable calculation	Shot detects	Shot non-detection
Above threshold	501	0
At or below threshold	0	939

The observable calculation uses the same threshold as the shot-based differential oracle and shares frozen QASM and release controls, while implementing its own parser and numerical operations.

Table S8 Breadth counts tests detecting a subject, conditional on the 70 test-catalogue-detectable subjects. Restricted unions below count detected files out of 120; omitted breadth bins have 0 subjects.

Breadth	2	4	5	6	7	8	9	10	11	12
Subjects	3	12	3	5	7	29	2	1	3	5

Test-catalogue restriction	Detected subjects
3-basis union at slot 0	61
3-basis union at slot 1	67
3-basis union at slot 2	66
3-basis union at slot 3	70
Z basis, all slots	62
X basis, all slots	46
Y basis, all slots	40
Z and X, all slots	70
Z and Y, all slots	70
X and Y, all slots	60

Table S9 Summed worker elapsed time over 120, 480 and 1,440 target invocations, respectively. Concurrent work is added, not converted to wall time; controls and suite construction are excluded.

Suite	Tests	Detections	Total time (s)	Time/detection (s)
T_B	1	50	0.451	0.0090
T_M	4	68	1.729	0.0254
Test catalogue	12	70	4.992	0.0713

Local notation for historical predicates.

Here $c(\mathcal{S})$ counts detections by a suite; $c_a(\mathcal{S})$ restricts that count to scope a. These symbols describe historical checks only. $c_{\text{base}} = c(T_B)$, $c_{\text{inh}} = c(T_M)$, $c_{\text{all}} = c(\mathcal{T})$, n_{sub} is the mutant denominator, $b_m = \sum_t D(m,t)$, and $c_Z = c(\{0,3,6,9\})$.

D_{shot} and D_{ref} are shot and exact detection indicators. Here $I_m(\mathcal{S}) = \mathbf{1}[\exists t \in \mathcal{S} : D(m,t) = 1]$ is a subject's detection indicator, and d_{mt} is the maximum absolute probability gap for subject m and test t. $\bar{b}$ is mean breadth among detected subjects. The local subscript P denotes the pooled result; v in the ordinal predicate indexes programs. Subscript 0 denotes the baseline-preserving comparator family; $\mathcal{C}_a$ contains all suites in the scope and comparator family indexed by a.

For each scope/family a, μ_a is the uniform mean, $N_a = |\{\mathcal{S} : c_a(\mathcal{S}) \geq c_{\text{inh,a}}\}|$, and $r_a = N_a/|\mathcal{C}_a|$, including ties and T_M.

$\mathcal{X}$ contains the reported baseline-preserving versus unconstrained and lineage-omission contrasts; the existential sensitivity rule requires a change somewhere, not

Table S10 Full forced-baseline comparison. The fixed T_M benchmark stays unchanged; for forced test a, G_a is separately reselected on development data with the original greedy rule and tie-break. Each comparator has 165 suites; $c(G_a)$ is its greedy-suite detection count.

Forced test a	Mean	$\geq c(T_M)$	G_a in addition order	$c(G_a)$	$\geq c(G_a)$
0	66.73	57	$[0, 11, 3, 1]$	68	57
1	65.84	73	$[1, 9, 0, 2]$	68	73
2	65.04	57	$[2, 9, 0, 1]$	68	57
3	67.68	102	$[3, 11, 0, 1]$	68	102
4	66.28	79	$[4, 9, 0, 1]$	70	53
5	65.46	69	$[5, 9, 0, 1]$	68	69
6	68.07	108	$[6, 11, 0, 1]$	69	97
7	66.22	79	$[7, 11, 3, 0]$	68	79
8	65.20	69	$[8, 10, 0, 1]$	65	109
9	68.99	145	$[9, 1, 0, 2]$	68	145
10	66.28	79	$[10, 9, 0, 1]$	70	53
11	66.65	103	$[11, 3, 0, 1]$	68	103

Both inclusive counts include ties. The fixed T_M belongs only to comparators requiring 1 of its own tests; all G_a belong to their respective comparators. Test $t = 3\ell + b$ decodes as slot ℓ and basis b: Z=0, X=1, Y=2. Only the test-0 baseline was part of the original procedure.

Table S11 Post-hoc structural suite families. Every suite has 4 tests and receives equal weight inside its listed family.

Baseline constraint	Family	T_M in set?	Suites	Mean	Min	Max
Retained	Basis-balanced	Yes	96	66.97	62	70
Retained	Input-balanced	No	27	67.11	62	70
Retained	Both	No	12	67.75	65	70
Unconstrained	Basis-balanced	Yes	288	68.17	62	70
Unconstrained	Input-balanced	No	81	65.62	40	70
Unconstrained	Both	No	36	68.47	65	70

Retained means test 0 is required; unconstrained means there is no baseline-preservation constraint. Basis-balanced covers Z, X and Y; input-balanced uses distinct slots; Both meets both constraints. The fixed T_M belongs to the basis-balanced families but neither input-balanced family nor either joint family.

Table S12 Post-hoc omissions of each lineage and both largest edit categories. The final columns count suites out of 165 in the fixed primary comparator; other counts refer to remaining files.

Omitted group	Files left	$c(T_M)$	Union	Mean	Excess	$> c(T_M)$	$= c(T_M)$	$\geq c(T_M)$
Bell lineage	60	34	34	33.76	0.24	0	155	155
Deutsch–Jozsa lineage	90	51	53	49.97	1.03	44	16	60
Grover lineage	90	51	53	49.73	1.27	44	13	57
Declared preserving	80	68	70	66.73	1.27	44	13	57
Replacements	80	30	32	29.48	0.52	44	37	81

Each row recomputes detection counts on the remaining files; the suite set is unchanged.

Table S13 Muskit program/input contract from the frozen source inventory. Gate calls are static before compilation: control count, then the range across supplied sources. No body measures or resets qubits.

Program	Qubits	Computation and effect of external prefix	Gate calls
CE	6	Conditional execution: $q_{0:2}$ control a multiqubit operation on $q_{3:5}$, followed by uncomputation. Prefix varies the first 2 controls; internal X/H/P preparation remains.	17; 11–18
LittleBV	6	Bernstein–Vazirani body: H/CZ/H copies the basis pattern of $q_{0:2}$ into $q_{3:5}$. Prefix varies only q_0, q_1; $q_2 = 0$.	9; 8–10
QRAM	7	Controlled-swap routing between $q_{3:4}$ and $q_{5:6}$ via address q_2, then swaps into $q_{0:1}$ and applies CX/X. Prefix varies the latter pair; address H/P/H remains.	9; 8–10

Index ranges are inclusive. Prefix slot ℓ applies X wherever the little-endian integer ℓ has a set bit, before the full body; idle classical registers are retained. Inputs are ordinal, without common application semantics.

Table S14 Planned and amended complete-row denominators. Exclusions for build errors were fixed before outcomes; the original planned-denominator analysis remains incomplete. Composition uses upstream filename categories.

Program	Planned	Complete	Arity	Index	Add	Remove	Replace
CE	458	446	12	0	330	12	104
LittleBV	343	333	0	10	245	9	79
QRAM	371	357	14	0	290	9	58
Total	1,172	1,136	26	10	865	30	241

Arity and Index count excluded sources by build-error type. Add/Remove/Replace counts use complete rows; no observed detection outcome determines an exclusion.

Table S15 Exposed QuCheck and outcome-blind Muskit complete-row results. $c_{\text{base}} = c(T_B)$, $c_{\text{inh}} = c(T_M)$ and $c_{\text{all}} = c(\mathcal{T})$ count detected subjects, where $c(\mathcal{S})$ counts detections by suite $\mathcal{S}$ and $\mathcal{T}$ is the test catalogue; inclusive fractions include ties and the policy.

Quantity	QuCheck	Muskit
Evaluation mutants	120	1,136
$c_{\text{base}} = c(T_B)$	50	529
$c_{\text{inh}} = c(T_M)$	68	596
c_{all}, test-catalogue union	70	621
$c_{\text{inh}} - c_{\text{base}}$, gain over baseline	18	67
$c_{\text{all}} - c_{\text{inh}}$, union minus inherited	2	25
Detected-only breadth mean; minimum	7.16; 2	6.44; 1
Z-only union	62	582
Primary suites	165	165
Uniform mean μ_0	66.73	591.71
Count $\geq c_{\text{inh}}$; inclusive fraction	57; 0.3455	81; 0.4909
$c_{\text{inh}} - \mu_0$; $\mu_0 - c_{\text{base}}$	1.27; 16.73	4.29; 62.71
Cell agreements; disagreements	1,440; 0	13,632; 0
Slot-union; test-catalogue-union agreements	480; 120	4,544; 1,136
Cell-minimum margin; non-detection clearance	0.089933; 0.056513	0.010474; 0.006563

The non-detection clearance is τ minus the maximum gap among non-detected subject–test cells. Both uniform means refer to the primary baseline-preserving comparator. Breadth averages only the c_{all} detected subjects. Observable agreement is conditional on shared construction and release controls. Operator taxonomies and input semantics differ between mutant sets.

Table S16 All Muskit complete-row lineages and omissions. The primary comparator retains test 0; the unconstrained comparator drops that constraint. Each inclusive suite count includes ties and the fixed policy.

Scope	n_{sub}	c_{base}	c_{inh}	c_{all}	μ_0	Primary count	μ	Unconstrained count
CE	446	191	228	233	223.86	44	209.05	86
LittleBV	333	186	195	207	196.31	109	190.62	369
QRAM	357	152	173	181	171.54	78	165.57	262
Without CE	690	338	368	388	367.85	94	356.19	298
Without LittleBV	803	343	401	414	395.40	68	374.62	166
Without QRAM	779	377	423	440	420.17	90	399.67	233

n_{sub} counts mutants; $c_{\text{base}} = c(T_B)$, $c_{\text{inh}} = c(T_M)$, $c_{\text{all}} = c(\mathcal{T})$. μ_0, μ are mean detections over baseline-preserving and unconstrained suites. The primary and unconstrained count columns count suites with $c(\mathcal{S}) \geq c_{\text{inh}}$, out of 165 and 495, respectively. Means are rounded for display; signs and verdicts use unrounded values. Every row retains the same T_B and T_M.

in every contrast. $\mathcal{X}_{C4}$ restricts comparator swaps to pooled and each omission, and pooled-to-omission comparisons to each family; individual-program reversals are outside its evaluated scope. The comparison allowance K is distinct from the contrast sets $\mathcal{X}$ and $\mathcal{X}_{C4}$.

Table S17 Mathematical fixed holdout checks and unchanged archived outcomes on the buildable-subject analysis. The planned-denominator outcomes remain INCOMPLETE.

Check	Predicate	Archived outcome
C1	$\forall m, t:\ D_{\text{shot}}(m,t) = D_{\text{ref}}(m,t)$, also after slot/test-catalogue OR reductions; $\nexists(m,t) : \lvert d_{mt} - \tau\rvert \le 10^{-10}$. Report positive and negative margins. With declared-equivalent set $E_\star$, specificity requires $\forall m \in E_\star : I_m(\mathcal{T}) = 0$.	Agreement REPLICATION; specificity UNAVAILABLE (labels absent).
C2	$0 < c_{\text{all}} < n_{\text{sub}}$; $\bar{b} > 1 \wedge \min_{m:b_m>0} b_m \ge 2$; $c_Z > c_{\text{all}}/2$. Assess components separately; the bundle is their conjunction.	Limited union and Z majority true; redundancy false; PARTIAL.
C3	$c_{\text{inh}} > c_{\text{base}} \wedge \mu_0 - c_{\text{base}} \ge (c_{\text{inh}} - c_{\text{base}})/2 \wedge N_0 > 1$. Sensitivity requires $\exists(a,b) \in \mathcal{X} : r_a \ne r_b$; equivalently, no rank difference across those contrasts fails this component. Mean differences are reported alongside ranks.	Raw-gain pattern REPLICATION; sensitivity REPLICATION.
C4	$(c_{\text{inh}} > c_{\text{base}} \wedge c_{\text{inh}} \le \mu_0)\ \vee\ [\exists(a,b) \in \mathcal{X}_{C4} : \text{sgn}(c_{\text{inh,a}} - \mu_a) \ne \text{sgn}(c_{\text{inh,b}} - \mu_b)]\ \vee\ [\exists m \in E_\star : I_m(\mathcal{T}) = 0]$.	NON_ REPLICATION in evaluated pooled/omission scope; label component UNAVAILABLE.
Ordinal position	$c_{\text{inh,P}} - \mu_{P,0} > 0 \wedge \forall v : c_{\text{inh},v} - \mu_{v,0} > 0$ gives consistent positive direction; positive pooled excess with a nonpositive program excess gives lineage limitation; otherwise no positive pooled direction.	LINEAGE_ LIMITED.

Interpreting the historical outcomes.

Pooled ranks $81/165 = 0.4909$ and $229/495 = 0.4626$ differ: the first uses baseline-preserving suites, the second all equal-budget suites. This change satisfies C3 without relying on mean changes. The frozen implementation compares rank/mean pairs and also reports program contrasts; the displayed rank change independently satisfies the protocol's rank criterion.

The protocol's comparator wording leaves that scope less explicit than its implementation; LittleBV's reversal is retained in Table S16 rather than used to revise the saved outcome.

The frozen C4 label clause asks whether an equivalent subject remains undetected, whereas C1 specificity requires that for every equivalent subject. Those clauses can both hold; their labels do not make them logical opposites. We preserve this historical predicate rather than repair it after outcomes; its label component was unavailable here.

The raw-gain/2 cutoff is conventional, not a practical effect threshold; a single alternative tie suffices for $N_0 > 1$. Exact protocol wording and implementation: Online Resource 2, `r3/holdout/freeze/PROTOCOL.md`, section C3/C4; Online Resource 2, `r3/holdout/code/analyze_holdout_a1.py`.

Contract-relative census and witness records.

The terminology key above applies to all following census tables. Muskit uses the executor contract where no program-semantic restriction is documented; superposition inputs are outside that contract. QuCheck C subjects agree within the program-semantic contract; teleportation agrees on its full allowed data subspace. Per-subject records: Online Resource 2, `r3/analysis/results/op_subjects.json`.

Table S18 Application-input audit of I subjects. A: admissible-input witness; C: contract-limited; P: phase-only with no admissible-input witness. The final column counts A subjects with an existing-statistic gap $> \tau$.

Program	I	A	C	P	A: $d > \tau$
QuCheck Deutsch–Jozsa	3	0	3	0	0
QuCheck Grover	3	0	3	0	0
QuCheck Teleportation	1	0	1	0	0
QuCheck Superdense coding	2	0	2	0	0
QuCheck pooled	9	0	9	0	0
Muskit CE	100	24	0	76	16
Muskit LittleBV	82	9	0	73	9
Muskit QRAM	129	97	0	32	43
Muskit pooled	311	130	0	181	68

Table S19 Files (behaviour groups) by program and diagnostic class: E, equivalent build; I, input-limited; O, summary-limited; T, threshold-limited; D, detected. Each of the 4 slots must match up to global phase within 10^{-9}. Groups describe build behaviour and are not independent failure mechanisms.

Program	E	I	O	T	D
QuCheck Deutsch–Jozsa	10 (1)	3 (3)	0 (0)	0 (0)	17 (16)
QuCheck Grover	10 (1)	3 (3)	0 (0)	0 (0)	17 (16)
QuCheck Teleportation	10 (1)	1 (1)	0 (0)	0 (0)	19 (17)
QuCheck Superdense coding	11 (1)	2 (2)	0 (0)	0 (0)	17 (15)
QuCheck pooled	41 (4)	9 (9)	0 (0)	0 (0)	70 (64)
Muskit CE	63 (1)	100 (53)	50 (31)	0 (0)	233 (152)
Muskit LittleBV	44 (1)	82 (52)	0 (0)	0 (0)	207 (108)
Muskit QRAM	47 (1)	129 (87)	0 (0)	0 (0)	181 (122)
Muskit pooled	154 (3)	311 (192)	50 (31)	0 (0)	621 (382)

Pooled totals: QuCheck 120 (77); Muskit 1,136 (608).

Source boundaries for Table S20.

CE O source boundaries count expanded source gate calls; each row gives a boundary and the number of O subjects edited there. Boundary 18 follows the original 18 gate calls.

Gate-call boundary	O subjects
2	5
5	2
6	10
16	10
17	2
18	21

Source lines, gate tokens, full IDs and grouping are archived in Online Resource 2, `r3/analysis/results/op_cross.csv`; Online Resource 2, `r3/analysis/results/op_subjects.csv`.

Provenance and the unused historical component.

The `2026-09-09` protocol used the label "S" for a planned QChecker static-attribution component in the historical workflow; it does not denote fresh-shot arm S. That component supplies no results to this article: the reported inputs and reductions contain differential measurements and suite selection, without QChecker warning-attribution outcomes.

Table S20 Program × class × edit type, counting files. Empty classes are shown explicitly; the T residual is 0 throughout. QuCheck preserving files belong to the declared-preserving operator; Muskit uses Add, Remove and Replace.

QuCheck					
Program	Class	Preserving	Insertion	Deletion	Replacement
Deutsch–Jozsa	E	10	0	0	0
Deutsch–Jozsa	I	0	2	0	1
Deutsch–Jozsa	O	0	0	0	0
Deutsch–Jozsa	T	0	0	0	0
Deutsch–Jozsa	D	0	6	2	9
Grover	E	10	0	0	0
Grover	I	0	3	0	0
Grover	O	0	0	0	0
Grover	T	0	0	0	0
Grover	D	0	6	1	10
Teleportation	E	10	0	0	0
Teleportation	I	0	1	0	0
Teleportation	O	0	0	0	0
Teleportation	T	0	0	0	0
Teleportation	D	0	8	1	10
Superdense coding	E	10	1	0	0
Superdense coding	I	0	1	0	1
Superdense coding	O	0	0	0	0
Superdense coding	T	0	0	0	0
Superdense coding	D	0	5	3	9

Muskit					
Program	Class	—	Add	Remove	Replace
CE	E	—	61	0	2
CE	I	—	93	1	6
CE	O	—	48	0	2
CE	T	—	0	0	0
CE	D	—	128	11	94
LittleBV	E	—	44	0	0
LittleBV	I	—	80	1	1
LittleBV	O	—	0	0	0
LittleBV	T	—	0	0	0
LittleBV	D	—	121	8	78
QRAM	E	—	46	0	1
QRAM	I	—	120	3	6
QRAM	O	—	0	0	0
QRAM	T	—	0	0	0
QRAM	D	—	124	6	51

Table S21 Nearest subject maximum to each boundary: separate value and absolute-distance rows. Maxima are within each subject over slots (and bases and existing statistics for d); nearest D entries minimize those subject maxima. Table S1 instead minimizes detected individual cells. These are gaps; a margin subtracts its stated threshold. Empty classes have no nearest subject; T is 0 in both mutant sets.

Mutant set/class	Measure	u	f	d vs. ϵ_o	d vs. τ
QuCheck E	Value	4.441×10^{-16}	1.11×10^{-15}	3.331×10^{-16}	3.331×10^{-16}
QuCheck E	Distance	10^{-9}	10^{-10}	9.997×10^{-13}	0.056513
QuCheck I	Value	0.076120	8.882×10^{-16}	6.106×10^{-16}	6.106×10^{-16}
QuCheck I	Distance	0.076120	10^{-10}	9.994×10^{-13}	0.056513
QuCheck D	Value	0.076120	0.146447	0.250000	0.250000
QuCheck D	Distance	0.076120	0.146447	0.250000	0.193487
Muskit E	Value	6.661×10^{-16}	1.332×10^{-15}	1.332×10^{-15}	1.332×10^{-15}
Muskit E	Distance	10^{-9}	10^{-10}	9.987×10^{-13}	0.056513
Muskit I	Value	0.076120	1.554×10^{-15}	1.443×10^{-15}	1.443×10^{-15}
Muskit I	Distance	0.076120	10^{-10}	9.986×10^{-13}	0.056513
Muskit O	Value	0.076120	0.146447	7.772×10^{-16}	7.772×10^{-16}
Muskit O	Distance	0.076120	0.146447	9.992×10^{-13}	0.056513
Muskit D	Value	0.008555	0.017037	0.103553	0.103553
Muskit D	Distance	0.008555	0.017037	0.103553	0.047040

Full IDs, full precision and nearest subjects on each side: Online Resource 2, `r3/analysis/results/op_nearest.csv`. Changing ϵ_u, ϵ_s or ϵ_o individually by $\times 10$ or $/10$ changes no class and introduces 0 tolerance conflicts. The shot threshold is sensitive to an increase:

Change	QuCheck E/I/O/T/D	Muskit E/I/O/T/D
$\tau/10$	41/9/0/0/70	154/311/50/0/621
$\tau \times 10$	41/9/0/45/25	154/311/50/358/263

Online Resource 2, `r3/analysis/results/op_summary.json`.
Distance means the absolute difference between the displayed value and the stated boundary.

Table S22 Historical CE O recovery by added pairwise parity observable and full-histogram total variation (TV), pooling slots. Saved shots: retrospective (q_2, q_3) reanalysis at the original unadjusted threshold; this historical result is not the adjusted all-pairs/TV audit in Table S31, outside the fixed protocol, with no new circuit execution. Raw positive gaps can include floating-point noise; resolved gaps exceed ϵ_o.

Basis	Pair $> \tau$	Pair > 0	Pair $> \epsilon_o$	TV $> \tau$	TV > 0	TV $> \epsilon_o$	Saved shots
Z	16	50	16	16	50	16	16
X	42	50	42	42	50	42	42
Y	45	50	45	45	50	45	45
Union	50	50	50	50	50	50	50

Kets here denote full-register basis integers. All 50 subjects have an exact pair (q_2, q_3) witness. `000f781fb6a1c07892cb390e`: CE, slot 0, registered $|0\rangle$, X: $P(q_2 \oplus q_3 = 1)$; exact gap 0.500000, sampled gap 0.498413 ($> \tau$); seed `1524242265652620292`. `0202259c4d869de58ca4a6aa`: LittleBV, slot 0, pre-build/application $|4\rangle$, Z: $P(q_5 = 1)$; exact gap 1.000000, sampled gap 1.000000 ($> \tau$); seed `5206019462096949305`. These are numpy multinomial simulations with 8,192 samples per role; only 1 CE subject has a seeded shot example. No new backend experiment was run. Under the program-semantic contract, QuCheck has no admissible input-change example. For saved-shot probability gaps $\hat{d}_{s\ell bj}$ (subject s, slot ℓ, basis b, statistic j), the saved-shot cell-minimum margin $\min_{\hat{d}>\tau} \hat{d} - \tau$ is 0.078741 at the original τ; the subject-maximum margin $\min_s \max_{\ell,b,j} \hat{d}_{s\ell bj} - \tau_{\text{pair}}$ is 0.089270 at the adjusted threshold. Their underlying gaps are 0.135254 and 0.147461. They use different cells and the same working pair, as Table S34 details; Online Resource 2, `r3/analysis/results/parity_shots.json` records each subject and input–basis cell. Saved probability arrays are normalized by their sums without rounding or clipping. For each per-simulation label archived in `op_repairs.json`, interpret its first 8 bytes of SHA256 as a big-endian unsigned integer; initialize `np.random.default_rng(seed)`, then draw the subject multinomial followed by the independent control multinomial. Online Resource 2, `r3/analysis/results/op_repairs.json`; Online Resource 2, `r3/analysis/results/op_extended.json`.

Table S23 Release controls checked against analytic expectations, followed by a separately implemented AST/numpy audit of Muskit source adaptation. Controls cover 36/36 registered builds. CE bits η, ζ are q_2, q_4; QRAM v, w are the bus bits before final CX/X. The pass tolerance is 10^{-10}; state comparisons allow global phase and probability comparisons use TV.

Control	Expected analytic property	Slots
Deutsch–Jozsa	Constant oracle gives 0 data; linear balanced oracle gives coefficient bits.	Matched 4/4
Grover	Marked-item success after the registered iteration is 1.	Matched 4/4
Teleportation	Reduced output fidelity to each registered data state is 1.	Matched 4/4
Superdense coding	Decoded data bits equal the registered classical message.	Matched 4/4
Fourier transform	Ascending-wire circuit equals $\Pi_{\text{bit}} F \Pi_{\text{bit}}$, with Π_{bit} the bit-reversal matrix and F the positive-phase Fourier matrix.	Matched 4/4
Phase estimation	The estimation register contains the bit-reversed eigenphase integer; the eigenstate is preserved.	Matched 4/4
LittleBV	Lower secret register is preserved; upper output equals the secret.	Matched 4/4
CE	Gate-derived: $q_2 = \eta$, $q_4 = \zeta$ have amplitude $e^{i\pi\zeta/4}/2$; the upper register starts at 2ζ and increments by $\eta \oplus \kappa$, $\kappa = 1 \oplus ((1 - q_0)q_1)$. The low bit flips and its neighbour is restored.	Matched 4/4
QRAM	Gate-derived: $HP(\pi/3)H$ gives address probabilities $3/4, 1/4$; controlled bank swaps and bus swap precede $(v, w) \mapsto (1 - v, w \oplus v)$.	Matched 4/4

In these closed forms only, $\eta, \zeta \in \{0, 1\}$ are CE bit values; QRAM v, w are the bus bits on q_0, q_1 immediately before the final CX/X gates. CE and QRAM expectations are closed forms derived from source gates, rather than external application specifications. Adapter passes: 4,556/4,556; builds containing `mct`: 1,652. Maximum overlap deviation: 8.882×10^{-16}; maximum phase-aligned entry error: 9.486×10^{-16}. Builds with global-phase magnitude above 10^{-10} radians: 748. Special paths cover `mct` to ancilla-free `mcx`, register reconstruction and order, static math/pi argument parsing, integer-input X prefixes, and frozen u/cx lowering. The separately implemented interpreter executes neither source code nor the adapter parser. Online Resource 2, `r3/analysis/results/op_controls.csv`; Online Resource 2, `r3/analysis/results/op_adapters.csv`.

Table S24 Representative I witnesses: all 9 QuCheck subjects and admissible-basis/phase-only examples from each Muskit program. A: admissible-input witness; C: contract-limited; P: phase-only with no admissible-input witness, with an outside superposition shown. Gap is the maximum existing-statistic probability difference over Z/X/Y for the chosen slot and state.

Program	Full source ID	Status	Slot	Pre-build state	d
Deutsch–Jozsa	`5540ccc9cc7a8858ca45917b`	C	0	$(\|0\rangle + \|1\rangle)/\sqrt{2}$	0.353553
Deutsch–Jozsa	`663b1d13e7211d91beabea6a`	C	0	$(\|0\rangle + \|1\rangle)/\sqrt{2}$	1.000000
Deutsch–Jozsa	`953255baf4b11ac82d9d926e`	C	0	$(\|0\rangle + \|4\rangle)/\sqrt{2}$	1.000000
Grover	`25a82a398b485a86f569e50d`	C	0	$\|4\rangle$	0.353553
Grover	`a1a342888e4642d0dd70331d`	C	0	$(\|0\rangle + \|4\rangle)/\sqrt{2}$	0.176777
Grover	`b1bfc15ed0e95b11faf7d48e`	C	0	$\|4\rangle$	1.000000
Teleportation	`c96ba53fb2d4a472293ae210`	C	0	$(\|0\rangle + \|2\rangle)/\sqrt{2}$	0.500000
Superdense coding	`34b06e6d4d982d18ca0b5812`	C	0	$(\|0\rangle + \|3\rangle)/\sqrt{2}$	1.000000
Superdense coding	`f22155415ba0e18c492cd073`	C	0	$(\|0\rangle + \|1\rangle)/\sqrt{2}$	0.353553
CE	`0fb76edbfd2109a521a8a3c5`	A	0	$\|33\rangle$	0.707107
CE	`063eec1e4b0e55fc02b06a85`	P	0	$(\|0\rangle + \|16\rangle)/\sqrt{2}$	1.000000
LittleBV	`0202259c4d869de58ca4a6aa`	A	0	$\|4\rangle$	1.000000
LittleBV	`02c2b95d3f7709f6a53b5139`	P	0	$(\|0\rangle + \|8\rangle)/\sqrt{2}$	0.353553
QRAM	`071d396bc01aa5c444ab5d2f`	A	0	$\|40\rangle$	1.000000
QRAM	`161b81a154c721aa20034709`	P	0	$(\|0\rangle + \|28\rangle)/\sqrt{2}$	0.433013

Integer kets use the complete register. For Muskit, application input is the pre-build integer XOR slot; the displayed A examples all use slot 0. C and P examples are outside the established input contract. The full I inventory comprises QuCheck A/C/P = 0/9/0 and Muskit A/C/P = 130/0/181. Each stored I row retains the full ID, chosen slot, witness, gap and legal-input audit: Online Resource 2, `r3/analysis/results/op_subjects.csv`; Online Resource 2, `r3/analysis/results/op_subjects.json`.

Table S25 Historical original identifiers; public copies may differ. Original hashes do not certify translated or summarized copies or external preregistration. Online Resource 2, `PAPER_SOURCE_MAP.json` maps original to distribution hashes and identifies translations, summaries and withheld originals.

r3/holdout/freeze/PROTOCOL.md
O: 552f606f486bb59b1c6a35f850b543ad9bd43bf187d6ac45d2b7cdcc75d9bc94
D: db279ca660f17cd6fb4ea5d3999c66e01254d077e975f9a4da3cff1d30da3818
r3/holdout/freeze_h2/CODE.sha256
O: 10dbb13df2cdd9e607b9a4d8b6465a2364b1fd731cbeea82f4e49d42730b0120
D: cf8fa279f083a3522263b14930fac6fd5871ff6cf92643511207a39fd0a6d7fc
r3/holdout/amendments/A1_STILLBORN.md
O: 1364d5d07382c3274bdb634f0cfd6af7b8d5247ecc1201d83eb0845aa47c05c3
D: 36eaf90b0fa3ea380336293ad0e30f4da31a5ad1c1655d5847934a83cf7f6b13
r3/holdout/amendments/A1b_NOTE.md
O: 1c60d086ae372161e2986801142d05fc866c013320ee94e6e38b064596a6bf99
D: 765983de0e2aaf8c4b62d7c338a9c97706e6c331d2c2d84f908b65aeff336e72
r3/holdout/amendments/A1.sha256
O: 40f0c491c93be7d2262c4551c9a898a44fa08ee8bcc02b6bb3772d0583e81863
r3/holdout/code/adapter.py
O: ce92386bf85b2dca94bd2afe5166299924ea072d8f2157db54985e4c3ce71b01
r3/holdout/code/run_holdout.py
O: 9dda23e5aa4462db987ca7bc62cbe1bad7b42817d1c1e4525aa6e8546731c7f7
r3/holdout/code/analyze_holdout.py
O: c5dc7bd1f5725116de67df266d065370f37ae045f1b09c8be7fda6f19144da4b
D: 02d2367b745f854f1d4944efaae7de27474e964279d8c1e8c1cbc2bc725a3119
r3/holdout/tests/test_holdout.py
O: d488c063eafd490b33bf94d7d1b4a1250299af289299e2a2bd42246991bb4b00
D: 602c916cd5da479c93cb6f524b3181cded99c63f26309ea5b104a87c169a15a1
r3/holdout/env/lock.txt
O: e4e53b873e00e3f17e8f0048798a8644a3b20410a1e326e79d7ea19fc8dfc80d
r3/holdout/env/ENV.json
O: 2bce5142432f13397bda7b9f9ed1922889d52d9ad40f2e80f42e1ab906d1a6b3
D: 9f0122eebefcb750a1bb64a196407dfc6272cd3105c8a93a3d9395c1681ad283
r3/holdout/env/requirements-hashed.txt
O: b9d6e4695251f5280d48167f56a14e9da24fc87357546f9de9fe0403d1d44cc4
r3/holdout/env/dry-inventory.json
O: eaf13ad50ebdebb77a2b5c9ac87cf3d522d1c37ffcdfd840422c29c0d48c4049
r3/holdout/env/dry-inventory-summary.json
O: ae62683460f95802d26ef50dd87845df4b0934d70ab8bada4e15ae149f3a820e
r3/holdout/env/tests.log
O: aa15318d27ad0c1f1f9cdccfee652bf865fe0d391884a7e329fd893bef2450c2
D: 0ef31f8aadeab8cbac22ec53bc85ab6f48311416a62d8bd02f2645f9a5915a16
r3/holdout/docs/H2_REVIEW.md
O: 81b8bde65aab944eb4973f94b488475458680de29a33f96e880726b59bec3545
D: 1261dd1e67d38be68c502532494485cac0a888e5feb7260ff5f5d6f581636309
r3/holdout/tests/fixtures/bell.py
O: d3de6f4c36924ea3367ad9bfceac283b85e33d83a724b8cbb0fb3b83692dc7a8
r3/holdout/tests/fixtures/gates.py
O: fe769e6d371d0d78ef582bc1614b710f18356c839e64cf92b0931eb2ab2a19ec
r3/holdout/tests/fixtures/ghz.py
O: 3834b6a73a96f6f14d7da11e561176371b39abaddb1d43e22cda11a268529844
r3/holdout/tests/fixtures/hst.py
O: ac54d63ec85b27da5cf49e929a4d67a65f9e7c9bb8f61601fd7e2d0fc3d74bc0
r3/holdout/tests/fixtures/mct.py
O: cd9e4147c5759f36f45550eab649dea640b6fc043db20c601fa460604c8ecaaa

O: historical original digest; D: public distribution digest where different. Protocol and amendments preceded Muskit outcomes; historical full manifest: Online Resource 2, `r3/holdout/freeze_h2/CODE.sha256`. Analysis code and QASM source digests: Online Resource 2, `r3/analysis/results/op_summary.json`.

Table S26 All repeated QuCheck QASM signatures. Each signature joins family and ordered raw build digests; this differs from unitary behaviour grouping.

1b12c262695016f4604db1962d9ada069e3f634b8515f122b9984d53d637efd0
b32f01a113a924a010321e78, eabd9c589e892233385cf5da
7013523df3a266ac38f199356897cd7a639c85bf69e00423259fb05ac245e735
1b536f03b36cffcfd1ffd620, 8261d973aef9451dd136cceb
76f80e2592936bdaa02d7585f4bf00063651c8699e62d30728c887c44644841b
5fd35e1068cadb4eac60892e, 70e8c573b8bcd6b9be24c3bd

Complete membership and file IDs: Online Resource 2, `r3/analysis/results/non_witness_summary.json`; collapsed suite outcomes: Online Resource 2, `r3/analysis/results/non_witness_summary.json`. Collapsed baseline/inherited suite/test-catalogue union: 49, 67, 68; mean 65.24, tail fraction 0.3091 (Online Resource 2, `r3/paper_numbers/generated_core_tables.md`, T10).

Table S27 Both 1-test CE detections. Columns are test IDs in order; these rows defeat the all-detected-subject redundancy condition.

Source ID	0	1	2	3	4	5	6	7	8	9	10	11
5c0a0b9c1546f2e25cfe40bd	0	0	0	0	0	0	0	0	0	1	0	0
3aeccae34edefa0bffee8946	0	0	0	0	0	0	1	0	0	0	0	0

5c0a0b9c1546f2e25cfe40bd: ExperimentalData/CE/Mutants/19AddGate_cswap_inGap_1_.py
3aeccae34edefa0bffee8946: ExperimentalData/CE/Mutants/227AddGate_ccx_inGap_12_.py
Complete rows: Online Resource 2, `r3/holdout/results/ANALYSIS_A1.json`.

Table S28 Muskit suite-time and census-cost scopes. Suite sums reuse 1 execution; census costs come from 1 later local analysis.

Recorded quantity	Value
Baseline-preserving suites	165
Target calls per suite	4,544
Target seconds: minimum / median / maximum	20.368 / 21.619 / 22.199
Inherited suite seconds / ascending rank	21.460 / 49
Maximum/minimum spread	9.0%
Census wall seconds / peak RSS (KiB)	22.21 / 401,912
Full analysis wall seconds / peak RSS (KiB)	26.22 / 584,648
Builds / maximum qubits	5,024 / 7

Suite time sums successful mutant `CELL` records' `backend.time_taken`; controls, starts, failures, construction, transport and scheduling are excluded. Every suite's full sum and detections: Online Resource 2, `r3/analysis/results/budget_sweep.json`, `muskit_target_time.suite_results`. Census includes input/witness search, CE extension, adapter checks and groups; full analysis adds analytic controls and output writing, excluding the final summary write, selftests and independent review. Peak RSS uses Linux `resource.getrusage` in KiB. Python `3.14.4`, NumPy `2.5.0`, `OPENBLAS_NUM_THREADS=1`; environment and scope: Online Resource 2, `r3/analysis/results/op_summary.json`, `cost`. Dense matrices use 16×4^n bytes each; caches and simultaneous slots increase peak memory beyond 1 matrix.

Table S29 Evidence and exposure. Chronology and complete protocol/code/amendment digests are retained in the supporting hash inventory.

Evidence layer	Fixed information, exposure and scope
QuCheck execution	Development determined the inherited suite. Comparator fixed after its evaluation raw gain, before added tests; original GPU binary/interpreter patch unpinned.
Record reduction	3,036 saved histograms rebuilt matrices and suite counts; same-node, same-seed record replay compares saved records, without fresh draws.
Muskit execution	Source rule, prefixes, sequence, checks, code and static exclusions fixed before Muskit outcomes, after QuCheck exposure; pinned CPU environment.
Numerical recomputation	Separate parser/gates on preserved builds; 36/36 analytic control checks and 4,556/4,556 source-to-unitary adapter checks.
Retrospective diagnosis	Unitary/state/input/observable census, detections, groups and budget sweep followed both outcomes; seeded test-change shots are seeded NumPy simulations.

QuCheck: Python `3.11`, Qiskit `1.2.4`, GPU Aer `0.15.1`; Muskit: Python `3.11.15`, same Qiskit/Aer, CPU statevector. Optimization level 0 and recorded seeds are fixed; manifests distinguish reduction, recomputation and original simulator execution.

Exploratory test changes and contract sensitivity

The following historical full-histogram TV and all-pairs parity arms use separate adjusted thresholds. The old TV threshold flags equivalent builds; applying the large TV threshold to existing statistics also loses some detections. The split-alpha Pearson comparison below replaces that arm as the main histogram comparison.

Table S30 Historical statistic families fixed before their calculations, after exposure to earlier CE findings.

Program	Qubits	Added pairs	L	τ_{pair}	τ_{TV}
CE	6	180	264	0.058191	0.115294
LittleBV	6	180	264	0.058191	0.115294
QRAM	7	252	348	0.058768	0.155316
Deutsch–Jozsa	3	36	84	0.056513	0.061821
Grover	3	36	84	0.056513	0.061821
Teleportation	3	36	84	0.056513	0.061821
Superdense coding	2	12	48	0.056513	0.056513

Each family covers 12 tests per subject. All pairs are fixed before this calculation, after exposure to earlier CE findings. The thresholds control each stated family per subject, not both arms or the complete mutant sets. Definition and event counts: Online Resource 2, `r3/analysis/results/rev_observables.json`.

Table S31 Historical conservative TV comparison, retained with its arm-design limitations.

Mutant set	Mode	Subjects	Original	Pairs	TV old	TV adj.	E FP old	D lost
QuCheck	Exact	120	70	70	70	70	0	0
QuCheck	Shots	120	70	70	70	70	0	0
Muskit	Exact	1,136	621	671	671	667	0	4
Muskit	Shots	1,136	621	671	904	668	65	3

Pairs means existing statistics plus all pairwise parity observables at the adjusted threshold. TV old uses the original threshold; TV adj. applies the histogram-adjusted threshold to that arm. Both adjusted arms flag 0 E subjects and no additional silent E/I subjects. E FP old is false detections among equivalent builds at the original TV threshold; D lost is original detections lost at the adjusted TV threshold. All O subjects are recovered by both adjusted arms. These saved control histograms are reused and do not estimate an independent false-positive rate. The historical TV arm applied its large threshold to existing statistics as well as TV; its QRAM losses are an arm-design artifact. The split-alpha Pearson comparison in Table S43 retains all original detections; independent control draws are calibrated in Table S44. Full class/program and control counts: Online Resource 2, `r3/analysis/results/rev_observables.json`.

Historical suite example and outcome records

The suite T_M is inherited from the exposed QuCheck development set; its raw gain is the increase over baseline detections, while excess is the difference from the uniform mean over equal-budget comparators. Its QuCheck raw gain decomposes as $18 = 16.73 + 1.27$; the full budget sweep describes these suites without evaluating a transferable selection method.

Worked example: a D subject missed by the inherited suite

The given `superdense_coding_m9.py` deletes `qc.z(0)` from message encoding, so the input-to-measurement path has a concrete missing operation. At slot 3, message 11, execution prepares a Bell pair, encodes the message, decodes, rotates into the selected basis, and samples both qubits. The release control decodes to both bits 1, whereas the mutant loses c_1 on q_0, so Z at test 9 exposes the difference while Y at test 11 hides it. Table S32 gives the sampled probabilities. The vector is $(\Pr[q_0 = 1], \Pr[q_1 = 1], \Pr[\text{odd parity}])$; $\hat{d}$ is the largest absolute difference between corresponding mutant and control components.

Table S32 Changing the measurement basis exposes the missing encoding gate in the illustrative Bell-lineage mutant.

Slot / basis	Sampled mutant probabilities	Sampled control probabilities	$\hat{d}$	Detects
3 / Z	$(0.0000, 1.0000, 1.0000)$	$(1.0000, 1.0000, 0.0000)$	1.000000	1
3 / Y	$(0.5056, 0.5009, 0.5057)$	$(0.5055, 0.4968, 0.5031)$	0.004028	0

The exact Z differences are $(1, 0, 1)$, while all exact Y differences are 0, matching the shot decisions against $\tau \approx 0.0565$. The fixed $T_M = \{0, 11, 3, 1\}$ misses this mutant; replacing its final X test with the slot-3 Z test gives $\{0, 11, 3, 9\}$. That equal-budget alternative detects 70 subjects, including both Bell files missed by T_M, though only this deletion is explained here. This example was chosen after evaluation; it does not explain or validate the earlier development-set choices.

Further numerical census examples.

LittleBV `Add102.py`, source ID `0202259c4d869de58ca4a6aa`, inserts $\mathrm{CZ}(q_2, q_5)$ and is compared with release control `19e8137ff9361a09de206867`. Across slots, $u = 0.500000$, $f = 0.000000$, $d = 2.22 \times 10^{-16}$: it is non-E but I. At slot 0, application input $|4\rangle$ yields infidelity 1 and Z gap $d = 1.000000$ on q_5, so it is A under both contracts. A seeded multinomial example gives $\hat{d} = 1.000000$; search-wide fresh-shot arm S flags it in 1,000/1,000 repetitions under either contract. Muskit executor arm B cannot reject by construction; its archived zero events are an implementation check. Slot-consistent B flags this subject in 1,000/1,000 repetitions. For contrast, CE source `000f781fb6a1c07892cb390e`, relative to control `0727e4d7dd62e11e9f4ff109`, has $u = 0.292893$, $f = 0.500000$, $d = 2.22 \times 10^{-16}$ and is O under both contracts.

Its registered X histogram at slot 0 changes, and generic Pearson testing recovers it; (q_2, q_3) explains the correlation, with exact gap 0.500000 and seeded multinomial gap 0.498413 (Online Resource 1, Table S22).

Source-tool outcome cross-tab.

`FAILED` is a literal outcome in Muskit's original test report, including `FAILED WITH P-Value nan`; the latter records an undefined p-value, not a valid rejection. Exact member and SHA`256` linkage gives 1,129 source subjects, with 7 unlinked subjects omitted without near-name substitution. Under executor, the linked E/C/O classes contain $154 + 0 + 50 = 204$ subjects, or 18.1%; all subjects in those classes are linked to such a label. The source-tool label and the wrapper class describe different tests and input domains; their agreement in total is not a source-tool false-positive rate or a revised mutation score. The full slot-consistent E/C/O classes sum to $154+171+50 = 374 + 1$ files; the linked cross-tab instead counts 374 (33.1%). Their difference is 1 unlinked E/C/O subject, excluded by the exact member/SHA`256` rule above; it does not change the census. QuCheck's full source-tool ratio is not derivable without its missing outcome-to-instance crosswalk (Table S55).

Complete exploratory reanalysis

The following analyses address contract sensitivity, histogram testing, independent specificity and native QuCheck property-based tests after earlier outcomes were known. All earlier observations and protocols remain preserved; these additions do not turn the historical holdout into independent confirmation.

Cell minima and subject maxima.

The saved-shot cell minimum is 0.135254. Its margin is 0.078741 at $\tau = 0.056513$ and 0.077063 at the adjusted pair threshold $\tau = 0.058191$. The minimum subject maximum is 0.147461. Its original-threshold margin is 0.090948 and its adjusted-threshold margin is 0.089270. The exact minimum subject maximum is 0.146447, with adjusted margin 0.088255.

The cell-minimum witness uses slot 3, basis Y and pair (q_2, q_3); the subject-maximum witness is a different subject, using slot 1, basis X and the same pair. The former subject has a larger gap, 0.155396, at slot 0 in Y. Thus the differing reported minima have different witnesses and aggregation orders. For every one of the 50 O subjects, this pair attains the maximum over all pairs and existing statistics. The other 14 pairs have saved-shot maximum 0.029175, below the original threshold.

Table S33 Reconciliation of different aggregations, denominators and units retained in the historical tables.

Quantity	Values	Interpretation
QuCheck detected gap	0.146447 / 0.250000	Individual-cell minimum / minimum of within-subject maxima; subtract the stated threshold only when reporting a margin.
Muskit detected gap	0.066987 / 0.103553	Individual-cell minimum / minimum of within-subject maxima; subtract the stated threshold only when reporting a margin.
Inclusive fractions	81/165 vs. 229/495	Baseline-preserving and unconstrained suites have different denominators.
DJ existing builds	124	Each of 30 mutants and 1 release control has 4 existing oracle builds; added oracles are a separate check.
Comparison allowance	117 / 120	$(12 + 1)(8 + 1)$ scalar comparisons; rounded upward conservatively.
Old contract subset	311 / 1,136	The old panel restricted I subjects only. The replacement revalidates every subject and legal slot.
A and cumulative test changes	130: 68/76/118	A establishes a state witness. Input / input plus pairs / input plus histogram need separate statistic checks.
Test-change scopes	50 / 8	Registered-input O recovery and extra A recovery after input search are different populations.
CE E units	63 / 1	Mutant files / behaviour groups; the second count is not an independent-mechanism count.
CE O source boundaries	50 total	Table S20 prints gate-call boundaries and subject counts as separate columns.
Development prefix	18, 38, 39, 39	Cumulative detections on 60 development subjects for the selected 1-, 2-, 3-, 4-test prefixes.
Gain decomposition	18 = 16.73 + 1.27	Raw gain above baseline = uniform mean gain above baseline + inherited-suite excess above that mean.

These distinctions change no original result. Pair gap/margin nesting is detailed in Tables S22 and S34. Full precision and selectors: Online Resource 2, `r3/analysis/results/rev2_reader.json`.

Table S34 Gap and margin are different quantities; cell minima and minima of subject maxima use different nesting.

Mode	Aggregation	Threshold	Gap	τ	Margin
Shots	Cell minimum	Original	0.135254	0.056513	0.078741
Shots	Cell minimum	Adjusted	0.135254	0.058191	0.077063
Shots	Subject maximum	Original	0.147461	0.056513	0.090948
Shots	Subject maximum	Adjusted	0.147461	0.058191	0.089270
Exact	Cell minimum	Original	0.146447	0.056513	0.089933
Exact	Cell minimum	Adjusted	0.146447	0.058191	0.088255
Exact	Subject maximum	Original	0.146447	0.056513	0.089933
Exact	Subject maximum	Adjusted	0.146447	0.058191	0.088255

The cell-minimum margin takes the smallest detected cell gap and subtracts its threshold; the subject-maximum margin first takes each recovered subject's maximum gap, then the smallest of those maxima, and subtracts its threshold. The pair (q_2, q_3) attains every subject maximum; existing statistics and the other pairs do not increase it. Explicit formulas appear in Table S22. Online Resource 2, `r3/analysis/results/rev2_margins_subjects.csv`.

Slot-inconsistent sensitivities.

CE lower fixes q_0–q_2 and QRAM lower fixes q_0, q_1. Both are explicitly slot-inconsistent: each excludes slots 1, 2 and 3, removing 9 registered tests per subject by construction. CE loses 4,014 subject–test cells and moves 14 D files into 3 A and 11 C; QRAM loses 3,213 cells and moves 19 D into 12 A and 7 C. These are excluded preparations, not newly discovered survival mechanisms. All per-program masks, legal/removed slots and exact transition IDs are in Online Resource 2, `r3/analysis/results/rev3_audit_masks.csv` and Online Resource 2, `r3/analysis/results/rev3_audit_D_transition_ids.csv`. Combined mask labels list CE, LittleBV and QRAM in that order; $q_{a:b} = 0$ fixes every wire from a through b inclusively. An asterisk marks a slot-inconsistent mask.

Whole-set workspace sensitivity.

Table S35 reclassifies all 1,136 Muskit subjects under 9 contracts: the executor contract; separate CE-lower, CE-upper, LittleBV-upper, QRAM-lower and QRAM-upper restrictions; and the joint CE-upper/LittleBV-upper/QRAM-lower, CE-upper/LittleBV-upper/QRAM-upper and CE-lower/LittleBV-upper/QRAM-lower restrictions. CE-lower fixes q_0 through q_2 to zero; CE-upper and LittleBV-upper fix every wire from q_3 upward. QRAM-lower fixes q_0, q_1; QRAM-upper fixes every wire from q_3 upward. A separate restriction affects only its named program; the other programs retain the executor contract.

For a fixed-wire mask W, a slot ℓ is admitted only if $\ell \,\&\, W = 0$. An admissible application basis string x also has zeros on those wires, and its before-build preparation is $x \oplus \ell$. Undoing a slot prefix does not make an excluded slot admissible. E has highest priority and remains fixed globally; D and O are reconsidered using admitted slots at the registered input, with the original threshold for D. T denotes a positive but subthreshold gap. Otherwise, search for A over all admissible basis inputs, then

P over their coherent span, and assign C if neither witness exists. Every contract has 0 T subjects.

Under the executor contract the counts are E=154, A=130, C=0, P=181, O=50 and D=621. Restrictions make 33 formerly D subjects silent: 14 in CE-lower and 19 in QRAM-lower. They comprise 15 new A subjects and 18 new C subjects. They remain silent at registered inputs; input search exposes the new A subjects. The full contract view includes these subjects, rather than reusing only the original silent cohort.

Table S39 separates each contract's own A population from the fixed original A cohort. For the executor contract, input search, addition of pairs, and cumulative addition of Pearson histograms give exact-law witnesses for 68, 76 and 118 of 130 subjects. The joint CE-upper/LittleBV-upper/QRAM-lower contract instead gives 62, 62 and 103 of its own 113 A subjects. With QRAM-upper in that combination, all 8 of 8 A subjects have exact-law witnesses at every stage. Changing denominators is not evidence of improved detection. These exact-distribution diagnostics do not control multiplicity over searched inputs or over the entire subject set.

Table S35 Whole-set Muskit census under all 9 input contracts; every row contains 1,136 subjects.

Input contract	E	A	C	P	O	D
Executor	154	130	0	181	50	621
CE $q_{0:2} = 0$*	154	131	79	118	47	607
CE $q_{3:5} = 0$	154	108	32	171	50	621
LittleBV $q_{3:5} = 0$	154	126	31	154	50	621
QRAM $q_{0:1} = 0$*	154	139	19	172	50	602
QRAM $q_{3:6} = 0$	154	34	108	169	50	621
Upper/upper/lower*	154	113	82	135	50	602
Slot-consistent	154	8	171	132	50	621
Lower/upper/lower*	154	136	129	82	47	588

Lower/upper name the wires held at 0: CE lower $q_0, \ldots, q_2$; CE upper and LittleBV upper $q_3, \ldots, q_5$; QRAM lower q_0, q_1; QRAM upper $q_3, \ldots, q_6$. Upper/lower labels name CE, LittleBV and QRAM explicitly; joint labels give CE / LittleBV / QRAM in order. A star marks excluded slots. All are sensitivity assumptions, not recovered application specifications. Only the named program is restricted in a single-program row. A slot ℓ is admitted only if its prepared basis string leaves the fixed wires zero. For an admitted slot, each searched application input x also leaves those wires zero; the before-build input is x XOR ℓ. E retains its full-build meaning. T and the registered measurement-basis-limited class are 0 in every row. CE lower and QRAM lower move 14 and 19 formerly D subjects into silent classes. Source: Online Resource 2, `r3/analysis/results/rev2_census.json`.

Table S36 Complete CE rows for the same 9-contract census.

Input contract	E	A	C	P	O	D
Executor	63	24	0	76	50	233
CE $q_{0:2} = 0$*	63	25	79	13	47	219
CE $q_{3:5} = 0$	63	2	32	66	50	233
LittleBV $q_{3:5} = 0$	63	24	0	76	50	233
QRAM $q_{0:1} = 0$*	63	24	0	76	50	233
QRAM $q_{3:6} = 0$	63	24	0	76	50	233
Upper/upper/lower*	63	2	32	66	50	233
Slot-consistent	63	2	32	66	50	233
Lower/upper/lower*	63	25	79	13	47	219

Lower/upper name the wires held at 0: CE lower $q_0, \ldots, q_2$; CE upper and LittleBV upper $q_3, \ldots, q_5$; QRAM lower q_0, q_1; QRAM upper $q_3, \ldots, q_6$. Upper/lower labels name CE, LittleBV and QRAM explicitly; joint labels give CE / LittleBV / QRAM in order. A star marks excluded slots. All are sensitivity assumptions, not recovered application specifications. Only the named program is restricted in a single-program row. A slot ℓ is admitted only if its prepared basis string leaves the fixed wires zero. For an admitted slot, each searched application input x also leaves those wires zero; the before-build input is x XOR ℓ. E retains its full-build meaning. Rows repeat when a restriction affects another program. Source: Online Resource 2, `r3/analysis/results/rev2_census_tables.csv`.

Table S37 Complete LittleBV rows for the same 9-contract census.

Input contract	E	A	C	P	O	D
Executor	44	9	0	73	0	207
CE $q_{0:2} = 0$*	44	9	0	73	0	207
CE $q_{3:5} = 0$	44	9	0	73	0	207
LittleBV $q_{3:5} = 0$	44	5	31	46	0	207
QRAM $q_{0:1} = 0$*	44	9	0	73	0	207
QRAM $q_{3:6} = 0$	44	9	0	73	0	207
Upper/upper/lower*	44	5	31	46	0	207
Slot-consistent	44	5	31	46	0	207
Lower/upper/lower*	44	5	31	46	0	207

Lower/upper name the wires held at 0: CE lower $q_0, \ldots, q_2$; CE upper and LittleBV upper $q_3, \ldots, q_5$; QRAM lower q_0, q_1; QRAM upper $q_3, \ldots, q_6$. Upper/lower labels name CE, LittleBV and QRAM explicitly; joint labels give CE / LittleBV / QRAM in order. A star marks excluded slots. All are sensitivity assumptions, not recovered application specifications. Only the named program is restricted in a single-program row. A slot ℓ is admitted only if its prepared basis string leaves the fixed wires zero. For an admitted slot, each searched application input x also leaves those wires zero; the before-build input is x XOR ℓ. E retains its full-build meaning. Rows repeat when a restriction affects another program. Source: Online Resource 2, `r3/analysis/results/rev2_census_tables.csv`.

Table S38 Complete QRAM rows for the same 9-contract census.

Input contract	E	A	C	P	O	D
Executor	47	97	0	32	0	181
CE $q_{0:2} = 0$*	47	97	0	32	0	181
CE $q_{3:5} = 0$	47	97	0	32	0	181
LittleBV $q_{3:5} = 0$	47	97	0	32	0	181
QRAM $q_{0:1} = 0$*	47	106	19	23	0	162
QRAM $q_{3:6} = 0$	47	1	108	20	0	181
Upper/upper/lower*	47	106	19	23	0	162
Slot-consistent	47	1	108	20	0	181
Lower/upper/lower*	47	106	19	23	0	162

Lower/upper name the wires held at 0: CE lower $q_0, \ldots, q_2$; CE upper and LittleBV upper $q_3, \ldots, q_5$; QRAM lower q_0, q_1; QRAM upper $q_3, \ldots, q_6$. Upper/lower labels name CE, LittleBV and QRAM explicitly; joint labels give CE / LittleBV / QRAM in order. A star marks excluded slots. All are sensitivity assumptions, not recovered application specifications. Only the named program is restricted in a single-program row. A slot ℓ is admitted only if its prepared basis string leaves the fixed wires zero. For an admitted slot, each searched application input x also leaves those wires zero; the before-build input is x XOR ℓ. E retains its full-build meaning. Rows repeat when a restriction affects another program. Source: Online Resource 2, `r3/analysis/results/rev2_census_tables.csv`.

Table S39 All contract-specific A detections versus the unchanged original A cohort (130 subjects).

Contract	Own A	Input	+Pairs	+Hist.	Old A eligible	Input	+Pairs	+Hist.
Executor	130	68	76	118	130	68	76	118
CE $q_{0:2} = 0$*	131	69	77	119	127	66	73	115
CE $q_{3:5} = 0$	108	54	54	96	108	54	54	96
LittleBV $q_{3:5} = 0$	126	64	72	114	126	64	72	114
QRAM $q_{0:1} = 0$*	139	80	88	129	127	68	76	117
QRAM $q_{3:6} = 0$	34	26	34	34	34	26	34	34
Upper/upper/lower*	113	62	62	103	101	50	50	91
Slot-consistent	8	8	8	8	8	8	8	8
Lower/upper/lower*	136	77	85	126	120	62	69	110

Left: each contract's newly classified A subjects. Right: fixed executor A cohort; eligible counts only members with an admissible basis-state difference. Each last column is cumulative split-alpha original-plus-pairs plus Pearson; original-plus-Pearson alone gives the same totals. Lower/upper name the wires held at 0: CE lower $q_0, \ldots, q_2$; CE upper and LittleBV upper $q_3, \ldots, q_5$; QRAM lower q_0, q_1; QRAM upper $q_3, \ldots, q_6$. Upper/lower labels name CE, LittleBV and QRAM explicitly; joint labels give CE / LittleBV / QRAM in order. A star marks excluded slots. All are sensitivity assumptions, not recovered application specifications. Only the named program is restricted in a single-program row. No new D occurs within the original A cohort. New silent subjects from executor D explain why own-A totals can increase under restrictions. Input searches are exploratory. Online Resource 2, `r3/analysis/results/rev2_census_repair_tables.csv`.

Mixed local measurement bases.

A Pauli string is a tensor product of I, X, Y and Z; I is the identity and weight counts non-identity factors. A homogeneous setting uses the same axis on all qubits; a mixed setting may use different axes.

All product settings in $\{X, Y, Z\}^n$ give exactly $3n+3^2\binom{n}{2}$ distinct Pauli observables of weight at most 2 when only marginals and pairwise parity observables are retained. Summing projectors on unused wires gives the identity, so their local axes cannot change these statistics. The corresponding probability gap is half the absolute Pauli-expectation gap.

Table S40 covers every original silent non-E subject: 370 in total, comprising 9 QuCheck subjects under the program-semantic contract and 361 Muskit subjects under the executor contract. Registered inputs and searches over admissible basis inputs are reported separately. At registered inputs, mixed statistics detect the O subjects but do not expose the input-limited subjects. This is an exact numerical search, with threshold comparisons, rather than a new mixed-basis shot experiment.

For QRAM input searches, mixed pairs raise pair-only detections from 40 to 43, but add 0 subjects to the executor-wide union of 76 detections from existing statistics and homogeneous pairs. The remaining 12 subjects have state differences at other inputs but identical weight-at-most-2 Pauli statistics. Changing only the local pair bases cannot expose them. P subjects instead agree up to global phase on every admissible basis input.

A separate exploratory amendment computed all 16,384 Pauli coefficients per fixed input without stopping at its first gap and found a mixed correlation of weight 4 for every residual subject, with parity-probability gap 0.866025. This is the minimum nonzero weight at each chosen, fixed witness input only; it is not a minimum over all inputs. These witnesses use the executor contract and do not establish admissibility under every workspace assumption. Measurement-basis-limited (M) means that different output states have identical full histograms in the registered measurement bases. The registered measurement-basis-limited count is 0; the residual QRAM subjects receive this additional diagnosis only after input search. The registered O subjects remain summary-limited because their homogeneous full histograms already differ.

Table S40 Original silent non-E cohort: all local-Pauli marginal and pair statistics, with registered inputs separated from input search.

Mutant set / program	Class	Subjects	Registered	Input search	Homogeneous TV $> \epsilon_o$
QuCheck	C	9	0	0	0
Muskit CE	A	24	0	24	24
Muskit LittleBV	A	9	0	9	9
Muskit QRAM	A	97	0	43	85
Muskit	P	181	0	0	0
Muskit CE	O	50	50	50	50
QRAM residual subset	A	12	0	0	0

Mixed columns use all Pauli strings of weight at most 2 and the adjusted mixed-family threshold; nonzero, original-threshold and homogeneous-pair-adjusted counts agree here. The residual row is a subset of QRAM A, not an extra group. This fixed original cohort contains 361 Muskit and 9 QuCheck subjects. Separate phase-only input sensitivity uses 1-qubit $|+\rangle$ inputs and the all-$|+\rangle$ input before each slot, outside the basis-input contract: it distinguishes 181/181 P subjects with existing statistics (Online Resource 2, `r3/analysis/results/rev_phase.json`). Newly silent subjects under restrictions are included in Table S41. Online Resource 2, `r3/analysis/results/rev2_mixed_subjects.csv`.

Table S41 Complete silent non-E Muskit populations under all contracts: mixed local-Pauli weight-at-most-two diagnostics.

Input contract	Silent non-E	Registered	Input search
Executor	361	50	126
CE $q_{0:2} = 0^*$	375	47	124
CE $q_{3:5} = 0$	361	50	104
LittleBV $q_{3:5} = 0$	361	50	122
QRAM $q_{0:1} = 0^*$	380	50	138
QRAM $q_{3:6} = 0$	361	50	84
$(q_{3:5}, q_{3:5}, q_{0:1}) = 0^*$	380	50	112
Slot-consistent	361	50	58
$(q_{0:2}, q_{3:5}, q_{0:1}) = 0^*$	394	47	132

Labels and denominators come from each new census, including the 33 newly silent executor-D subjects (15 A, 18 C). The full view contains 6,668 subject–contract–scope rows, including both input scopes. Restrictions create no new registered mixed detection; the surviving O subjects remain summary-limited. Online Resource 2, `r3/analysis/results/rev2_mixed_contracts.csv`.

Table S42 All 12 QRAM residuals have weight-4 mixed-correlation witnesses at the fixed inputs shown.

Subject ID prefix	Application input	Pauli string	Min. weight	Parity gap
`235d4fbe`	54	`XXIIXIY`	4	0.866025
`3e8a5fd9`	28	`XXIIXIY`	4	0.866025
`85a92824`	28	`XXIIXIY`	4	0.866025
`85cf7c69`	52	`XXIIXIY`	4	0.866025
`a725fc13`	112	`IXIIXXY`	4	0.866025
`bdd29863`	9	`IXIIXXY`	4	0.866025
`c58fe0f4`	92	`IXIIXXY`	4	0.866025
`cbd4670b`	8	`IXIIXXY`	4	0.866025
`cc8024de`	88	`IXIIXXY`	4	0.866025
`e04a08bb`	28	`XXIIXIY`	4	0.866025
`e5d0af69`	92	`IXIIXXY`	4	0.866025
`f91ab35a`	8	`IXIIXXY`	4	0.866025

Every row uses slot 0 and the executor contract. Pauli strings run from q_6 to q_0, with I the identity. The local setting is `XXZZXZY` for `XXIIXIY` and `ZXZZXXY` for `IXIIXXY`; identity wires are completed with Z. The weight minimum is at that fixed input, not across every input. All marginal/pair gaps and homogeneous histogram gaps remain numerically 0 over the allowed basis-input search. The all-Pauli expansion was an exploratory amendment after the weight-at-most-2 negative result. Full IDs and tensor/rotation cross-checks: Online Resource 2, `r3/analysis/results/rev2_higherweight.csv`.

Split-alpha Pearson comparison.

For observed subject/control integer counts a_z, c_z in output bin z from independent samples of equal size $N = 8{,}192$, use the actual two-sample Pearson statistic

$$\mathcal{B} = \{z : a_z + c_z > 0\}, \quad X^2 = \sum_{z\in\mathcal{B}} \frac{(a_z - c_z)^2}{a_z + c_z},$$
$$\nu = |\mathcal{B}| - 1, \qquad\qquad p = \Pr\{\chi^2_\nu \geq X^2\}.$$

Only pooled-zero bins are removed; bins are not merged. If $\nu = 0$, set $p = 1$. With $\alpha = 0.001$, allocate $\alpha/2$ to existing statistics and $\alpha/2$ to the 12 histogram comparisons, rejecting a histogram only when $p < \alpha/(2 \cdot 12)$. The existing-statistic threshold is

$$\tau_{\text{split}} = \sqrt{\frac{2\log(4K/(\alpha/2))}{N}}, \qquad K = \max\{120, 12(n+1)\}.$$

Here K is the comparison allowance. All supplied builds have $n \leq 9$, so $12(n+1) \leq 120$ and this maximum equals the registered $K = 120$. Detection is the union of a strict gap exceedance and a histogram rejection. In the cumulative test-change comparison including pairs, replace the allowance by $\max\{120, 12[n + 1 + \binom{n}{2}]\}$ and retain the same alpha split. No histogram total-variation threshold is applied to a marginal.

The exact-distribution arm uses deterministic pseudo-counts Np_z and Nq_z after removing numerical roundoff zeros and renormalizing. It is a diagnostic, not an

observed hypothesis test or a power estimate. Pearson tails are asymptotic: cells with an expected count $(a_z + c_z)/2$ below 5 occur in 64/15,072 exact cells and 899/15,072 saved-shot cells. Sparse bins therefore limit finite-sample error guarantees.

Table S43 gives identical subject decisions for the exact and saved-shot arms: detections rise from 691 to 741 among 1,256, adding the 50 O subjects and losing 0 existing detections. Observed E detections are 0/195. This replaces the earlier conservative total-variation comparison as the principal histogram comparison; that historical comparison remains recorded.

Table S43 Split-alpha Pearson comparison: exact pseudo-count and saved-shot subject decisions agree.

Mutant set	Class	Subjects	Existing	Split existing	Pearson	Union
QuCheck	E	41	0	0	0	0
QuCheck	C	9	0	0	0	0
QuCheck	D	70	70	70	70	70
Muskit	E	154	0	0	0	0
Muskit	A	130	0	0	0	0
Muskit	P	181	0	0	0	0
Muskit	O	50	0	0	50	50
Muskit	D	621	621	621	621	621

Only the O class gains detections; no existing detection is lost, and saved-shot E detections remain 0/195. Exact pseudo-counts Np, Nq are diagnostic, not observations or power estimates. Sparse expected-bin cells (expected count below 5): exact 64/15,072, saved shots 899/15,072. Pearson uses an asymptotic approximation without pooling sparse bins. Online Resource 2, `r3/analysis/results/rev2_histogram_cells.csv`.

Independent-draw multinomial calibration.

Table S44 uses 1,000 repetitions for each of 195 E subjects and every test, totaling 2,340,000 mutant–control sample pairs. Each pair contains independent draws $A \sim \text{Multinomial}(8{,}192, p)$ and $B \sim \text{Multinomial}(8{,}192, q)$ from the respective archived exact laws. Draws are independent across subjects, test cells and repetitions; controls are never reused. Alternative comparisons use the same draw pairs. Streams use seed 20,261,101, stable subject digests and separate cell and role streams with NumPy SeedSequence and `PCG64`; each repetition index spans the entire subject set.

For Pearson alone and for the split-alpha union, per-subject false-positive counts range from 0 to 3 per 1,000 repetitions. There are 75 events among 195,000 subject repetitions, involving 62 subjects. The set-wide false-positive rate counts repetitions with any false positive: 71/1,000, or 0.071000, with Wilson 95% interval $[0.056670, 0.088614]$. Alpha was not divided across the subjects. Per-subject error control at α is not simultaneous control across the subject set; sparse-bin approximation also remains relevant.

The existing-statistic, adjusted-pair and adjusted-total-variation comparisons each have 0/1,000 set-wide events; a zero-event one-sided exact 95% upper bound is

0.00299125, not proof of zero error probability. The unadjusted total-variation diagnostic has 1,000/1,000 set-wide events. Individual subject intervals are not simultaneous intervals. All rates condition on archived noiseless laws and this subject set, rather than hardware or unseen programs.

Original draws used NumPy 2.3.5; verification fixed the archived NumPy 1.26.4 environment and checked stored draw checksums and events. Multinomial replay can differ across NumPy versions even with the same seed, so reproducing these events requires the recorded implementation and version.

Table S44 Independent-draw specificity: set-wide false-positive events in 1,000 independent repetitions.

Statistic family	QuCheck	Muskit	All subjects	Wilson low	Wilson high
Existing statistics	0	0	0	0.000000	0.003827
Existing + pairs, adjusted	0	0	0	0.000000	0.003827
Existing + TV, adjusted	0	0	0	0.000000	0.003827
Existing + TV, original threshold	0	1,000	1,000	0.996173	1.000000
Split-alpha Pearson union	13	58	71	0.056670	0.088614

Each repetition includes all 195 E subjects and 12 tests per subject; the denominator in every count column is 1,000. A set-wide event means at least one E subject is flagged. Zero-event arms have one-sided exact 95% upper bound 0.00299125. Per-subject error control does not add a correction across all 195 subjects. Intervals condition on the archived noiseless exact laws. Online Resource 2, `r3/analysis/results/rev2_calibration_summary.json`.

Native QuCheck PBT.

Native QuCheck property-based testing (PBT) preserves the archived generators, preconditions and assertions, separately from the fixed wrapper. Its exploratory protocol was written and hashed before execution: 9 properties, 4 accepted inputs per property, seed 20,261,101, $N = 8{,}192$ per measurement and native $\alpha = 0.01$. Native Fisher comparisons, Holm–Bonferroni adjustment and most-frequent checks were retained. The first attempt disabled native optimization and produced 93 `KeyError` failures when duplicate circuits lost measurement identifiers. Exceptions were not counted as detections or native non-detections.

A repair protocol was separately written and hashed before the corrected execution. It restored the native default `run_optimization=True` for circuit deduplication and measurement mapping, retaining native transpilation optimization level 0. Source code, properties, generators, inputs, preconditions, seed and statistical rules were unchanged. These are failed and corrected attempts within the same study, not independent repetitions. The first attempt retains count archives for 31 completed cases and error logs for the 93 failures; its wrapper did not save the failures' raw counts, leaving that evidence gap.

Table S45 reports only the corrected execution: 68/120 detections, agreement with the test catalogue for 116/120, E detections 0/41, release-control detections 0/4, and 0 execution errors. Agreement compares recorded native property-failure outcomes

with the test-catalogue union; it does not independently validate native significance calculations.

Native generator size parameters span Deutsch–Jozsa 3–10 qubits and Grover 4–7 qubits, rather than being restricted to the fixed wrapper's dimensions. E/C/D describe the fixed builds and are not equivalence claims across these generator ranges. Including controls, the corrected run used 3,255 measurement circuits and 26,664,960 shots, with 12–48 circuits per subject. Only shots per measurement match; total budgets are unmatched. Although corrected count archives are retained, their dumps lack the assertion-to-circuit mapping needed to recompute native p-values independently.

The failed attempt logged 4,464 measurement circuits and 36,569,088 shots, separately from the corrected budget above. Every algorithm family has 17/30 native detections. The offline CPU environment used Python 3.11.15, Qiskit 1.2.4, Aer 0.15.1, NumPy 1.26.4 and SciPy 1.17.1; all 296 archived Python source files compared byte-identically.

Base protocol SHA256: `182bf1f841d9fde4db012e9da9e13f859ad3dff93049b5fbf669ab8ff2786622`.

Repair protocol SHA256: `c54470f8f4f74e40d8833fcb260c82c8ff02d5996596273c1b94661fdb1b0faf`.

Table S45 Corrected native QuCheck PBT run, grouped by the fixed-wrapper census class.

Class / population	Subjects	Detected	Native non-detections	Error
E	41	0	41	0
C	9	1	8	0
D	70	67	3	0
All mutants	120	68	52	0
Release controls	4	0	4	0

Native PBT detects 68/120 mutants and agrees with the test catalogue on 116/120. Native-only: Grover m8; test-catalogue-only: Grover m11, teleportation m9 and m13. E/C/D classify fixed wrapper builds and do not claim equivalence across native generator sizes. Total circuit budgets differ. Online Resource 2, `r3/analysis/results/rev2_native.json`.

Separately implemented checks and preserved evidence.

A separate raw-QASM parser, gate engine, count decoder and gamma-tail calculation checked 30,144 histogram cells and 15,252 distinct archived histograms; multinomial verification checked all 4,680 cell-role streams and their event arrays. The final reanalysis self-check preserved 13,689 protected original files and all 496 earlier numerical values. These checks support the stored observations and arithmetic; native assertion p-values remain outside the independently reconstructed evidence because the saved dumps lack their assertion-to-circuit map. Reproduction requires the archived sampling environment, as described in Online Resource 2, `r3/analysis/`

`REV2_ENVIRONMENT.md`; source observations and analysis outputs remain separate from rendered tables.

Slot-consistent contracts and search-wide fresh-shot evidence

The executor contract admits all preparable basis inputs. Defined after exposure within the audited masks, the slot-consistent contract fixes CE and LittleBV q_3–q_5 and QRAM q_3–q_6 just after slot preparation, before the program body, preserving every slot and test. It fixes no wire varied by a slot or documented preparation at that boundary; the masks are assumptions, not recovered semantic specifications. Body-internal coherent gates are distinct from external preparation. Coherent phase probes can violate even these masks and remain outside both main contracts.

Table S46 Full Muskit contract sensitivity: files (behaviour groups).

Mask	E	A	C	P	O	D
Executor	154 (3)	130 (95)	0 (0)	181 (97)	50 (31)	621 (382)
CE $q_{0:2} = 0$*	154 (3)	131 (96)	79 (37)	118 (73)	47 (29)	607 (370)
CE $q_{3:5} = 0$	154 (3)	108 (76)	32 (24)	171 (92)	50 (31)	621 (382)
LittleBV $q_{3:5} = 0$	154 (3)	126 (91)	31 (20)	154 (81)	50 (31)	621 (382)
QRAM $q_{0:1} = 0$*	154 (3)	139 (101)	19 (18)	172 (89)	50 (31)	602 (366)
QRAM $q_{3:6} = 0$	154 (3)	34 (30)	108 (72)	169 (90)	50 (31)	621 (382)
Upper/upper/lower*	154 (3)	113 (78)	82 (62)	135 (68)	50 (31)	602 (366)
Slot-consistent	154 (3)	8 (7)	171 (116)	132 (69)	50 (31)	621 (382)
Lower/upper/lower*	154 (3)	136 (98)	129 (75)	82 (49)	47 (29)	588 (354)

Joint labels give CE / LittleBV / QRAM in order; upper and lower wire ranges are defined in the text. A star marks excluded slots. All rows retain the original 608 groups; T and M are 0. Upper means CE/LittleBV q_3–q_5 or QRAM q_3–q_6 fixed zero; lower means CE q_0–q_2 or QRAM q_0–q_1. Lower masks are slot-inconsistent. The slot-consistent row fixes CE and LittleBV q_3–q_5 and QRAM q_3–q_6 to zero at the external application-preparation boundary; it is an assumption, not an established specification. Per-program and subject rows: Online Resource 2, `r3/analysis/results/rev3_contract_census.csv` and Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S47 Complete CE contract census: files (groups).

Mask	E	A	C	P	O	D
Executor	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
CE $q_{0:2} = 0$*	63 (1)	25 (22)	79 (37)	13 (8)	47 (29)	219 (140)
CE $q_{3:5} = 0$	63 (1)	2 (2)	32 (24)	66 (27)	50 (31)	233 (152)
LittleBV $q_{3:5} = 0$	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
QRAM $q_{0:1} = 0$*	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
QRAM $q_{3:6} = 0$	63 (1)	24 (21)	0 (0)	76 (32)	50 (31)	233 (152)
Upper/upper/lower*	63 (1)	2 (2)	32 (24)	66 (27)	50 (31)	233 (152)
Slot-consistent	63 (1)	2 (2)	32 (24)	66 (27)	50 (31)	233 (152)
Lower/upper/lower*	63 (1)	25 (22)	79 (37)	13 (8)	47 (29)	219 (140)

Joint labels give CE / LittleBV / QRAM in order; upper and lower wire ranges are defined in the text. A star marks excluded slots. The same groups are retained under every mask. Fixed-wire definitions and slot-inconsistency are stated in the surrounding text. Full source IDs: Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S48 Complete LittleBV contract census: files (groups).

Mask	E	A	C	P	O	D
Executor	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
CE $q_{0:2} = 0$*	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
CE $q_{3:5} = 0$	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
LittleBV $q_{3:5} = 0$	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)
QRAM $q_{0:1} = 0$*	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
QRAM $q_{3:6} = 0$	44 (1)	9 (8)	0 (0)	73 (44)	0 (0)	207 (108)
Upper/upper/lower*	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)
Slot-consistent	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)
Lower/upper/lower*	44 (1)	5 (4)	31 (20)	46 (28)	0 (0)	207 (108)

Joint labels give CE / LittleBV / QRAM in order; upper and lower wire ranges are defined in the text. A star marks excluded slots. The same groups are retained under every mask. Fixed-wire definitions and slot-inconsistency are stated in the surrounding text. Full source IDs: Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S49 Complete QRAM contract census: files (groups).

Mask	E	A	C	P	O	D
Executor	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
CE $q_{0:2} = 0$*	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
CE $q_{3:5} = 0$	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
LittleBV $q_{3:5} = 0$	47 (1)	97 (66)	0 (0)	32 (21)	0 (0)	181 (122)
QRAM $q_{0:1} = 0$*	47 (1)	106 (72)	19 (18)	23 (13)	0 (0)	162 (106)
QRAM $q_{3:6} = 0$	47 (1)	1 (1)	108 (72)	20 (14)	0 (0)	181 (122)
Upper/upper/lower*	47 (1)	106 (72)	19 (18)	23 (13)	0 (0)	162 (106)
Slot-consistent	47 (1)	1 (1)	108 (72)	20 (14)	0 (0)	181 (122)
Lower/upper/lower*	47 (1)	106 (72)	19 (18)	23 (13)	0 (0)	162 (106)

Joint labels give CE / LittleBV / QRAM in order; upper and lower wire ranges are defined in the text. A star marks excluded slots. The same groups are retained under every mask. Fixed-wire definitions and slot-inconsistency are stated in the surrounding text. Full source IDs: Online Resource 2, `r3/analysis/results/rev3_contract_subjects.csv`.

Table S50 Exact-law A witness search for all Muskit contracts: files (groups).

Mask	Own A	Input	+ pairs	+ histogram
Executor	130 (95)	68 (58)	76 (65)	118 (89)
CE $q_{0:2} = 0$*	131 (96)	69 (59)	77 (66)	119 (90)
CE $q_{3:5} = 0$	108 (76)	54 (46)	54 (46)	96 (70)
LittleBV $q_{3:5} = 0$	126 (91)	64 (54)	72 (61)	114 (85)
QRAM $q_{0:1} = 0$*	139 (101)	80 (67)	88 (74)	129 (97)
QRAM $q_{3:6} = 0$	34 (30)	26 (23)	34 (30)	34 (30)
Upper/upper/lower*	113 (78)	62 (51)	62 (51)	103 (74)
Slot-consistent	8 (7)	8 (7)	8 (7)	8 (7)
Lower/upper/lower*	136 (98)	77 (64)	85 (71)	126 (94)

Joint labels give CE / LittleBV / QRAM in order; upper and lower wire ranges are defined in the text. A star marks excluded slots. These are deterministic numerical witnesses. Every scenario and cohort, including original A and newly silent subjects: Online Resource 2, `r3/analysis/results/rev3_audit_all_ladder_groups.csv`.

Fresh-shot allocation and scopes.

A role means subject or release control, and a rung is a cumulative family in the nested search. Arm S gives each elementary test 8,192 samples per role. Arm B divides the test-catalogue budget $12 \times 8{,}192 = 98{,}304$ per role (196,608 including control) over the same full search, using floor allocation and a one-shot remainder in fixed order. For H histograms, $L = H[n + 1 + \binom{n}{2}]$ counts scalar tests and $J = L + H$ counts elementary tests; scalar comparison allowance is $K = L$. The Input, + pairs and + histogram rungs share one search-wide per-subject $\alpha = 0.001$; half covers all scalar comparisons and half all histograms, without reallocation or renewed alpha at earlier rungs. Scalar samples are independent binomial pushforwards of full multinomial laws; histogram tests use independent full-law multinomial draws. Every searched test is sampled, without stopping or sharing controls. For CE/LittleBV executor $J = 17{,}664$, $N = 5$–6 in B; QRAM $J = 46{,}080$, $N = 2$–3. Slot-consistent searches use $J = 2{,}208$, $N = 44$–45 and $J = 2{,}880$, $N = 34$–35 respectively. The supplied protocol is Online Resource 2, `r3/rev3_evidence/PROTOCOL.md`, SHA256 `2aa69018f25f1395997d0695248ae874e9ec7270ba54b78a03c24037b30ce65e`.

Muskit executor B has $N \leq 6$, so $\tau_{\text{scalar}}(K, N) > 1$ and $X^2 \leq 2N \leq 2 \cdot 6$ cannot attain $p < \alpha/(2H)$, as derived in the manuscript. Its zero detections are an analytical design consequence; the archived simulation is retained only as an implementation check. All elementary tests have separate draws; sharing bitstrings across statistics at one circuit setting would give a different allocation, which was not evaluated. Its coordinate amendment uses direct prebuild columns for QuCheck and application input XOR slot for Muskit. These simulations add no circuit/backend execution. Each probability has 1,000 repetitions; groups use the fixed lexicographically smallest source-ID representative, never any-member OR. Group counts are descriptive, not independent mechanisms. Pearson removes pooled-zero bins without sparse-bin pooling; its asymptotic p-values do not guarantee exact finite-sample alpha. The search-wide adjustment covers one subject's whole search, not all 195 E files together.

Table S51 Executor: Muskit and QuCheck fresh-shot search: A files (groups), mean A file probability and any-E file (group) frequency.

Contract	Scope	Arm and rung	A detections	Mean A	Any E
Executor	Muskit	S: input	68 (58)	0.523077	0.000 (0.000)
Executor	Muskit	S: + pairs	76 (65)	0.584615	0.000 (0.000)
Executor	Muskit	S: + histogram	118 (89)	0.907731	0.052 (0.001)
Executor	Muskit	B: input	0 (0)	0.000000	0.000 (0.000)
Executor	Muskit	B: + pairs	0 (0)	0.000000	0.000 (0.000)
Executor	Muskit	B: + histogram	0 (0)	0.000000	0.000 (0.000)
Executor	QuCheck	S: input	2 (2)	1.000000	0.000 (0.000)
Executor	QuCheck	S: + pairs	2 (2)	1.000000	0.000 (0.000)
Executor	QuCheck	S: + histogram	2 (2)	1.000000	0.019 (0.004)
Executor	QuCheck	B: input	1 (1)	0.579000	0.000 (0.000)
Executor	QuCheck	B: + pairs	1 (1)	0.579000	0.000 (0.000)
Executor	QuCheck	B: + histogram	2 (2)	1.000000	0.013 (0.002)

Counts at $\hat{p} \geq 0.95$ and $\hat{p} \geq 0.5$ agree. All rates have 1,000 repetitions. Groups use the fixed smallest-source-ID representative, without any-member OR. Subject intervals and pointwise Wilson/Clopper–Pearson intervals for every scope are in Online Resource 2, `r3/analysis/results/rev3_ladder_summary.json` and Online Resource 2, `r3/analysis/results/rev3_ladder_subjects.csv`; neither interval family is simultaneous.

Table S52 Executor pooled and slot-consistent Muskit fresh-shot search: A files (groups), mean A file probability and any-E file (group) frequency.

Contract	Scope	Arm and rung	A detections	Mean A	Any E
Executor	ALL	S: input	70 (60)	0.530303	0.000 (0.000)
Executor	ALL	S: + pairs	78 (67)	0.590909	0.000 (0.000)
Executor	ALL	S: + histogram	120 (91)	0.909129	0.071 (0.005)
Executor	ALL	B: input	1 (1)	0.008773	0.000 (0.000)
Executor	ALL	B: + pairs	1 (1)	0.008773	0.000 (0.000)
Executor	ALL	B: + histogram	2 (2)	0.015152	0.013 (0.002)
Slot-consistent	Muskit	S: input	8 (7)	1.000000	0.000 (0.000)
Slot-consistent	Muskit	S: + pairs	8 (7)	1.000000	0.000 (0.000)
Slot-consistent	Muskit	S: + histogram	8 (7)	1.000000	0.059 (0.001)
Slot-consistent	Muskit	B: input	6 (5)	0.750250	0.000 (0.000)
Slot-consistent	Muskit	B: + pairs	8 (7)	1.000000	0.000 (0.000)
Slot-consistent	Muskit	B: + histogram	8 (7)	1.000000	0.002 (0.000)

Counts at $\hat{p} \geq 0.95$ and $\hat{p} \geq 0.5$ agree. All rates have 1,000 repetitions. Groups use the fixed smallest-source-ID representative, without any-member OR. Subject intervals and pointwise Wilson/Clopper–Pearson intervals for every scope are in Online Resource 2, `r3/analysis/results/rev3_ladder_summary.json` and Online Resource 2, `r3/analysis/results/rev3_ladder_subjects.csv`; neither interval family is simultaneous.

Table S53 Slot-consistent: QuCheck and pooled fresh-shot search: A files (groups), mean A file probability and any-E file (group) frequency.

Contract	Scope	Arm and rung	A detections	Mean A	Any E
Slot-consistent	QuCheck	S: input	2 (2)	1.000000	0.000 (0.000)
Slot-consistent	QuCheck	S: + pairs	2 (2)	1.000000	0.000 (0.000)
Slot-consistent	QuCheck	S: + histogram	2 (2)	1.000000	0.010 (0.001)
Slot-consistent	QuCheck	B: input	1 (1)	0.568000	0.000 (0.000)
Slot-consistent	QuCheck	B: + pairs	1 (1)	0.568000	0.000 (0.000)
Slot-consistent	QuCheck	B: + histogram	2 (2)	1.000000	0.014 (0.001)
Slot-consistent	ALL	S: input	10 (9)	1.000000	0.000 (0.000)
Slot-consistent	ALL	S: + pairs	10 (9)	1.000000	0.000 (0.000)
Slot-consistent	ALL	S: + histogram	10 (9)	1.000000	0.069 (0.002)
Slot-consistent	ALL	B: input	7 (6)	0.713800	0.000 (0.000)
Slot-consistent	ALL	B: + pairs	9 (8)	0.913600	0.000 (0.000)
Slot-consistent	ALL	B: + histogram	10 (9)	1.000000	0.016 (0.001)

Counts at $\hat{p} \geq 0.95$ and $\hat{p} \geq 0.5$ agree. All rates have 1,000 repetitions. Groups use the fixed smallest-source-ID representative, without any-member OR. Subject intervals and pointwise Wilson/Clopper–Pearson intervals for every scope are in Online Resource 2, `r3/analysis/results/rev3_ladder_summary.json` and Online Resource 2, `r3/analysis/results/rev3_ladder_subjects.csv`; neither interval family is simultaneous.

Table S54 Histogram-rung E specificity: individual mean versus any-E frequency.

Contract	Arm	Unit	E units	Mean $\hat{p}$	Any E	Wilson interval
Executor	S	files	195	0.00037949	0.071	[0.056670, 0.088614]
Executor	S	groups	7	0.00071429	0.005	[0.002138, 0.011651]
Executor	B	files	195	0.00006667	0.013	[0.007613, 0.022114]
Executor	B	groups	7	0.00028571	0.002	[0.000549, 0.007263]
Slot-consistent	S	files	195	0.00036923	0.069	[0.054884, 0.086415]
Slot-consistent	S	groups	7	0.00028571	0.002	[0.000549, 0.007263]
Slot-consistent	B	files	195	0.00008205	0.016	[0.009872, 0.025832]
Slot-consistent	B	groups	7	0.00014286	0.001	[0.000177, 0.005643]

Both earlier rungs have 0 mean and 0 any-E events in every scope. Every rung has 0 E units at probability thresholds 0.95 and 0.5. These pointwise Monte Carlo intervals do not provide simultaneous coverage. Exact Clopper–Pearson intervals and all subgroup scopes remain in Online Resource 2, `r3/analysis/results/rev3_ladder_summary.json`.

Pair-free O recovery and higher-weight audit.

The generic split-alpha Pearson family uses fixed bitstring bins, without pair selection, and recovers all 50 O files (31 groups) in exact laws and saved shots. It was designed after archive and CE-result exposure; its fixed family is not a prospective discovery claim. The pair (q_2, q_3) explains the correlation, without being required by the histogram test. The higher-weight audit computed all 16,384 coefficients per fixed input and used no first-gap stop; fallback count is 0. Its minimum weight is conditional on that input, distinct from the negative check over all 128 basis inputs, 4 slots and 210 weight-at-most-2 Pauli statistics. The exhaustive space, ordering and tie rule remain in Online Resource 2, `r3/analysis/results/rev3_audit_Q4.csv`.

Table S55 Source-tool outcome and original-binary dependency audit.

Evidence	Interpretation
Muskit source failures	1,129 exact member/SHA256 links; executor E/C/O 204 (18.1%); slot-consistent E/C/O 374 (33.1%). Reported failure includes `P-Value nan`. 7 unlinked subjects omitted, without near-name substitution.
QuCheck source ratio	Not derivable: partial notebook rows do not supply the full source outcome and generated-instance crosswalk.
CPU reconstruction	Saved QASM/source census, contract witnesses, stored-count reduction and CPU probability sampling do not require the original QuCheck GPU binary.
Native records	Saved native outcomes can be audited. Their missing assertion-to-circuit map prevents independent reconstruction of native p-values; a fresh native execution is a different claim.
Original execution	Generating original QuCheck shot records anew depends on the unpinned binary/interpreter environment. Historical runtime is a recorded observation, not a fresh-cost guarantee.
Missing observations	Unavailable historical resampling records cannot be reconstructed by obtaining a binary.
Locator inventory	1,312 source locations from the current manuscript, Online Resource 1 and historical source tables; repeated statements are retained, and unused rows are marked. The binding file records exact audited source hashes.

The cross-tab is exploratory, conditional on supplied-wrapper classes, and is neither a source-tool false-positive rate nor a revised mutation score. Exact IDs, hashes, scope and reconstruction limits: Online Resource 2, `r3/analysis/results/rev3_audit_Q5.json`, Online Resource 2, `r3/current_inventory/claims.csv`, Online Resource 2, `r3/current_inventory/Q6.json` and Online Resource 2, `r3/current_inventory/binding.json`.

Table S56 Every fixed-wrapper C subject and its recorded native outcome.

File	ID prefix	Qubits	Native	Consequence
Deutsch–Jozsa m2	`5540ccc9`	3	non-detection	Native range unproved
Deutsch–Jozsa m16	`663b1d13`	3	non-detection	Native range unproved
Deutsch–Jozsa m12	`953255ba`	3	non-detection	Native range unproved
Grover m14	`25a82a39`	3	non-detection	Native range unproved
Grover m10	`a1a34288`	3	non-detection	Native range unproved
Grover m8	`b1bfc15e`	3	detected	Native counterexample
Teleportation m8	`c96ba53f`	3	non-detection	Native range unproved
Superdense coding m19	`34b06e6d`	2	non-detection	Native range unproved
Superdense coding m14	`f2215541`	2	non-detection	Native range unproved

Deutsch–Jozsa and Grover use fixed 3-qubit builds, teleportation 3 and superdense coding 2. Native Grover m8 separates outside that fixed instantiation. Other non-detections do not prove C over native ranges. Full generator ranges, seeds, source digests, count dimensions, properties and protocol hashes: Online Resource 2, `r3/analysis/results/rev3_audit_native_C.csv`.

Details of the inherited-suite audit

Dependence and multiplicity

Every alternative detection beyond QuCheck's T_M lies in Bell, which has 2 times either other lineage's file weight. Without Bell, both inherited suite and union detect 34 subjects; no suite exceeds this count (Online Resource 1, Table S12). Equal lineage weights give detected shares of 56.67% for T_M, 55.82% for the uniform mean and 57.78% for the union, including silent files. These shares weight lineages equally; raw shares weight files equally.

Grouping identical ordered per-slot QASM build-digest sequences within families leaves 117 subjects, with baseline/inherited suite/union counts 49, 67, 68, uniform mean 65.24 and tail fraction 0.3091. These artifact groups differ from unitary behaviour groups (Online Resource 1, Table S26); neither grouping establishes independent faults.

Calls, recorded time and census cost

Muskit's 165 baseline-preserving suites each use 4,544 target calls, but accumulated backend times span 20.368–22.199 seconds, a 9.0% spread relative to the minimum. The median is 21.619 seconds, while T_M takes 21.460 seconds, ranking 49 of 165 from shortest to longest (Online Resource 1, Table S28). These single-run sums exclude controls, construction, transport, scheduling and repeated-run effects; QuCheck worker times have a broader scope (Online Resource 1, Table S9).

The census used 22.21 seconds and 401,912 KiB peak resident memory for 5,024 builds of at most 7 qubits, including witnesses, adapter checks and grouping. Including control checks and output writing gives 26.22 seconds and 584,648 KiB; these are single-run observations. Dense complex128 storage grows as 16×4^n bytes per unitary, and equal target calls do not imply equal end-to-end budgets (Online Resource 1, Table S28).